

Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

- (1) Ligand coordination and substitution ✓
(配体的配位和取代)
- (2) Oxidative addition and reductive elimination ✓
(氧化加成和还原消除)
- (3) Insertion and elimination (de-insertion) reactions
(插入反应和消除(反插入)反应)
- (4) Electrophilic and Nucleophilic Addition and Abstraction
(亲电和亲核加成与攫取)
(Reactions on the complexed ligands 和金属结合的配体的反应)

Help us to understand and predict the metal-catalyzed or mediated reactions.

Be aware:

- Classified in terms of reaction patterns, not mechanism !!
- An elementary reaction may go through different mechanisms.

Chapter 7. Insertion and Elimination

Outline

A. Introduction

B. CO insertion reactions

C. Insertion reactions involving alkene and alkyne

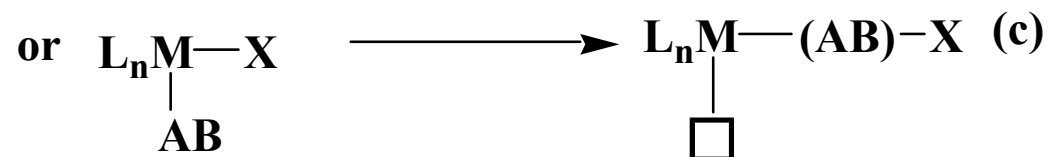
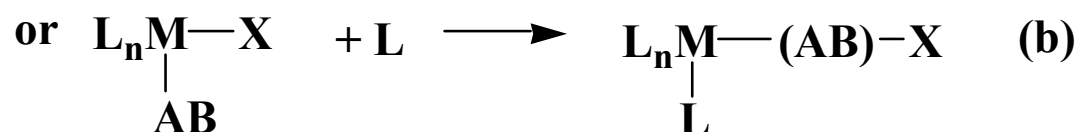
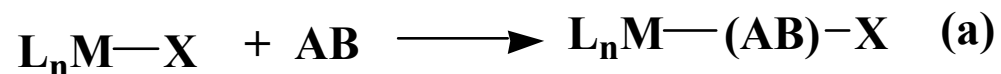
D. α , β , γ and δ elimination reactions

References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, **2014**, Chapter 6.
2. *Organometallic Chemistry*, G. O. Spessard, G. L. G. L. Miessler, Prentice-Hall: New Jersey, **1997**, Chapter 8.
3. *Advanced Inorganic Chemistry*, F. A. Cotton, G. Wilkinson, John Wiley & Sons, 5th ed. **1988**, Chapter 27, 1186.
4. *Organotransition Metal Chemistry*, Akio Yamamoto, **1986**, 246-272.

A. Introduction

Insertion reactions refer to the following processes.



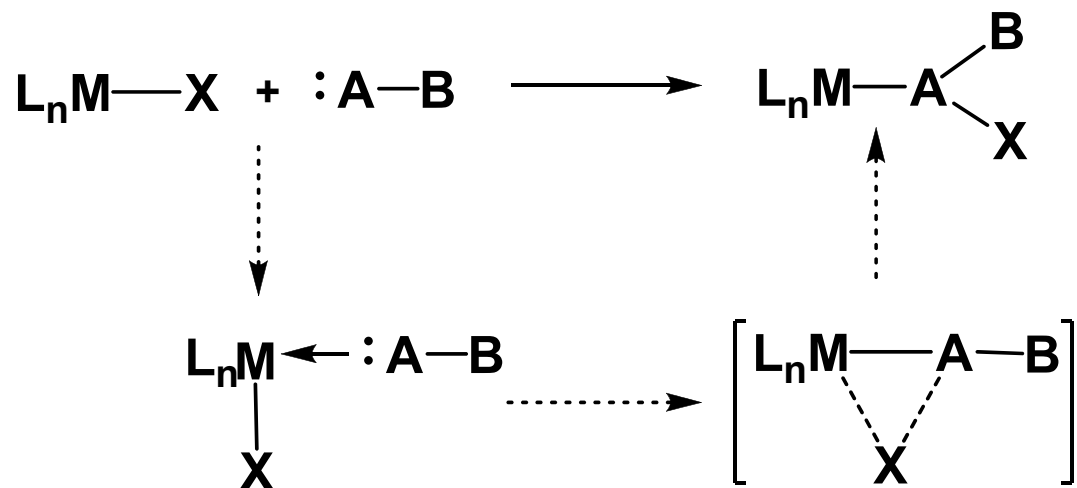
 may be involved in (a) and (b)

Common types of insertion reactions:

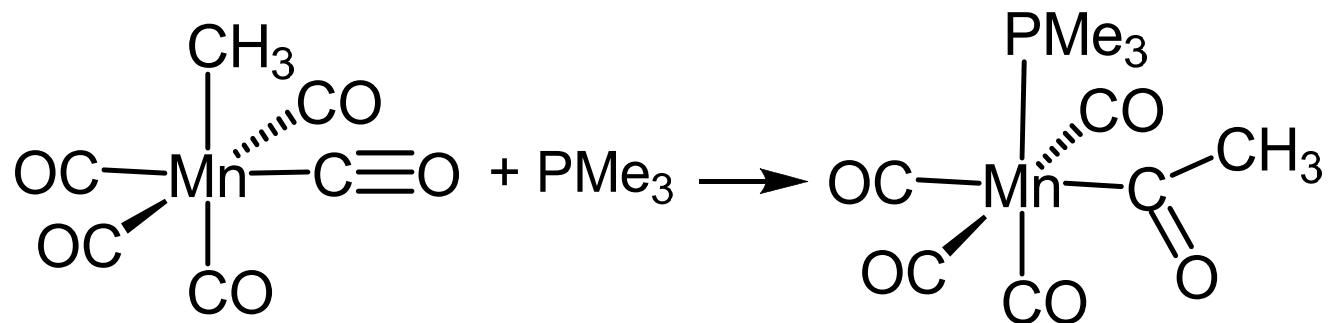
- 1). 1,1-insertion
- 2). 1,2-insertion
- 3). 1,3-insertion
- 4). 1,4-insertion

Common types of insertion reactions:

1). 1,1-insertion

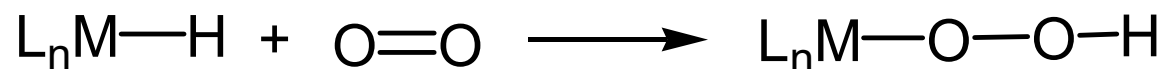


e.g.

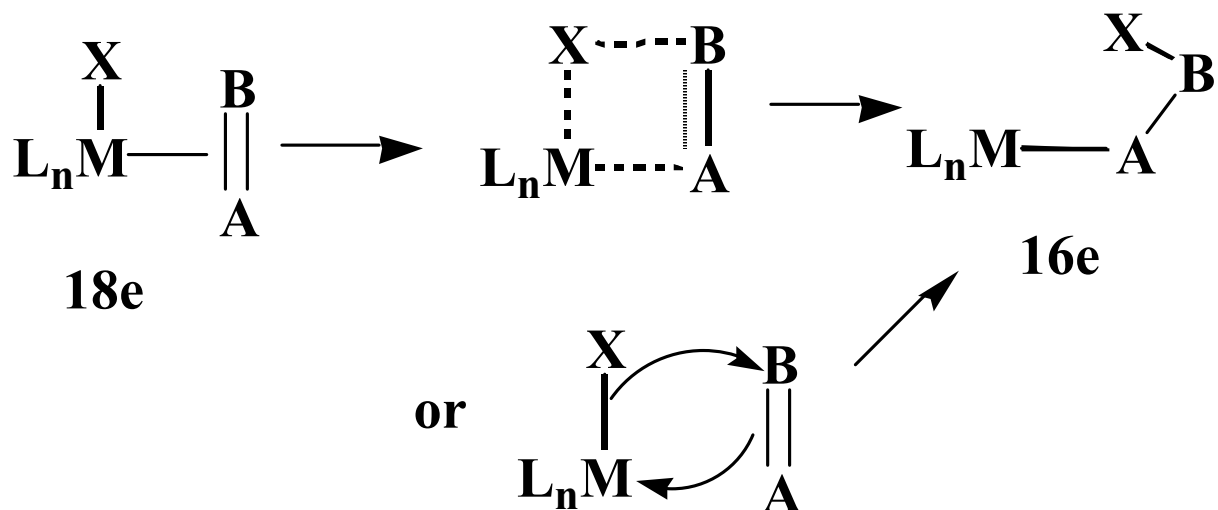


Common types of insertion reactions:

2). 1,2-insertion

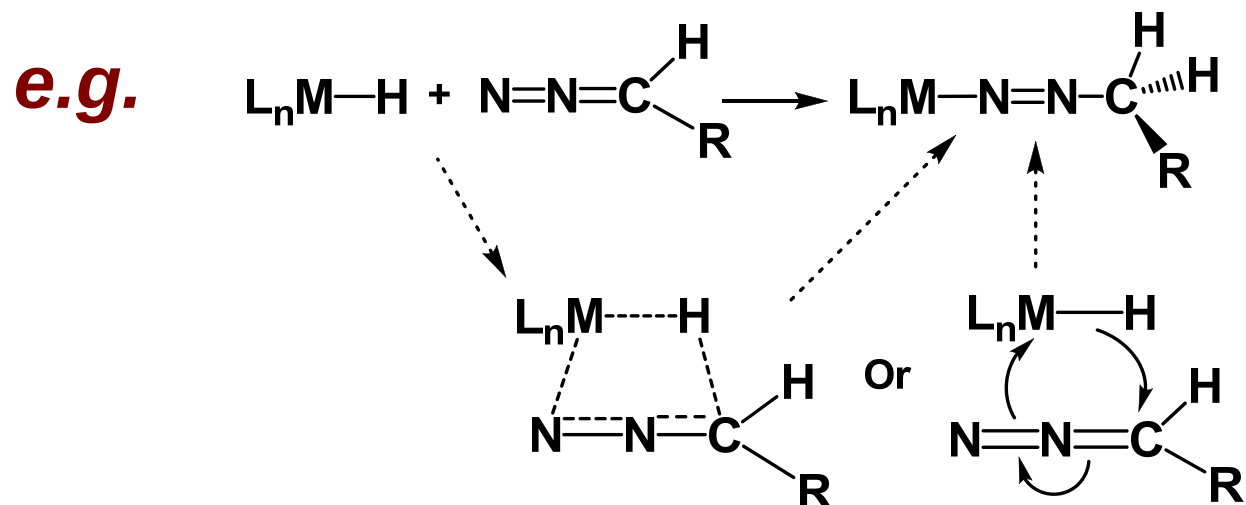


Often involve the following steps

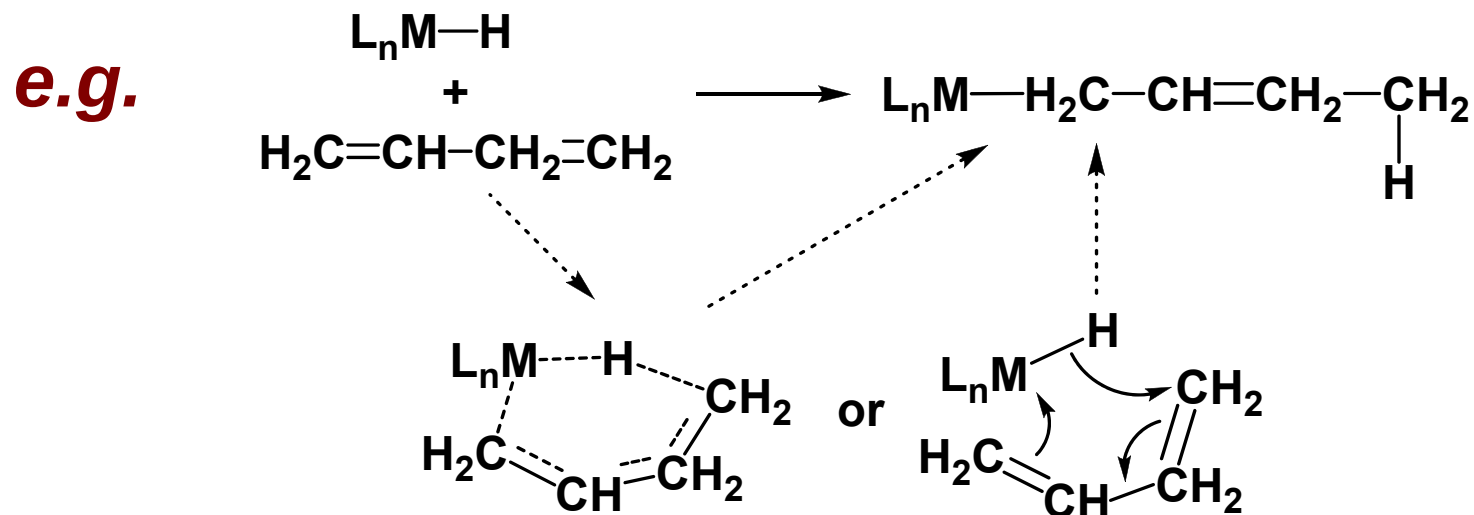


Common types of insertion reactions:

3). 1,3-insertion

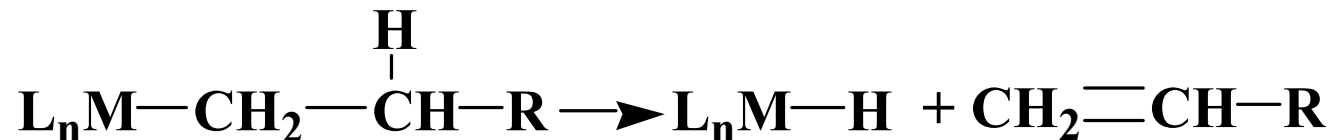
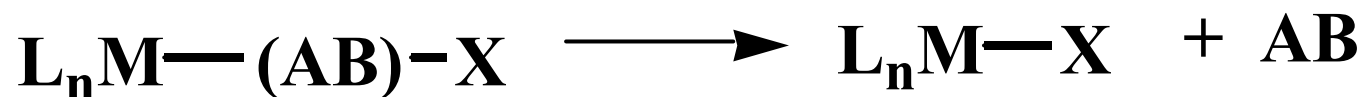


4). 1,4-insertion

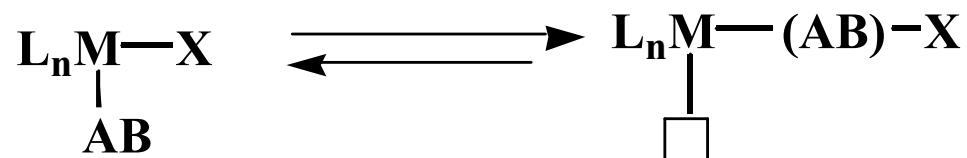


Elimination reaction:

Elimination (or de-insertion or extrusion) reactions are just reverse of insertion reactions.



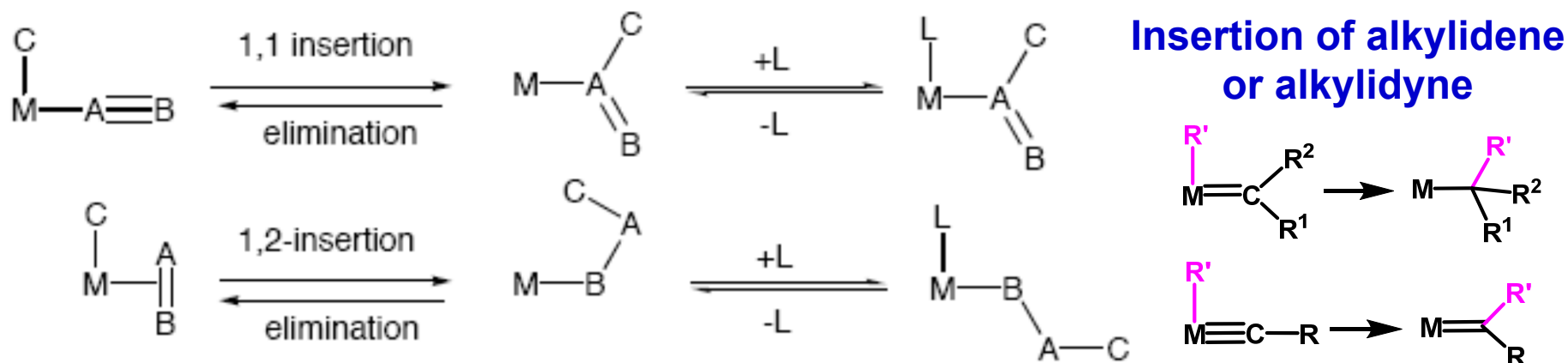
In principle:



If $\Delta G > 0$

If $\Delta G < 0$

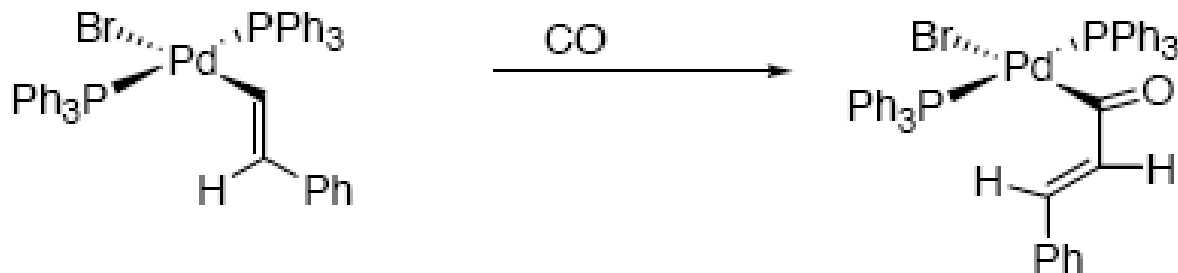
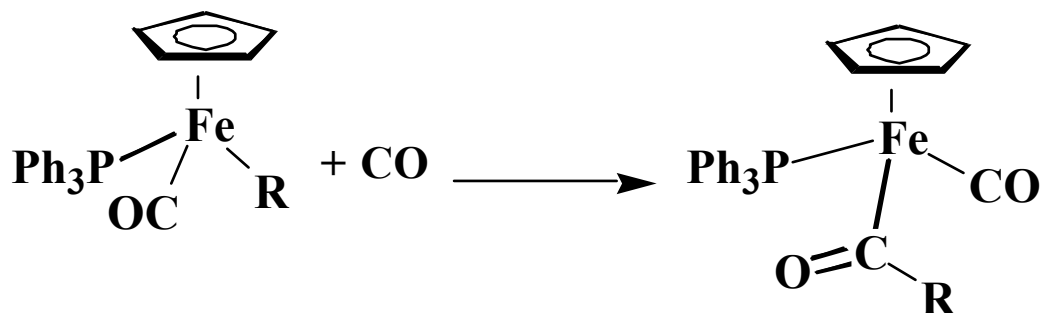
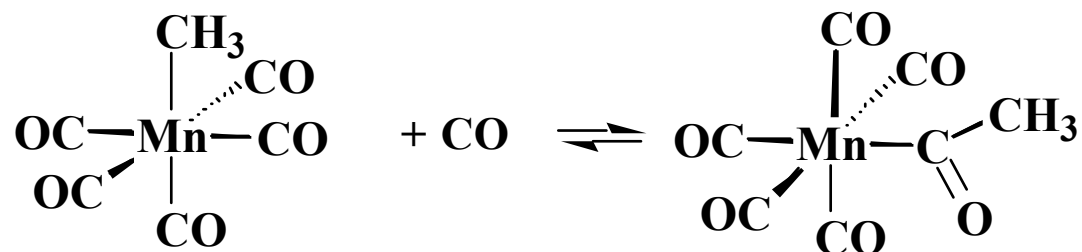
General features for insertion and elimination reactions



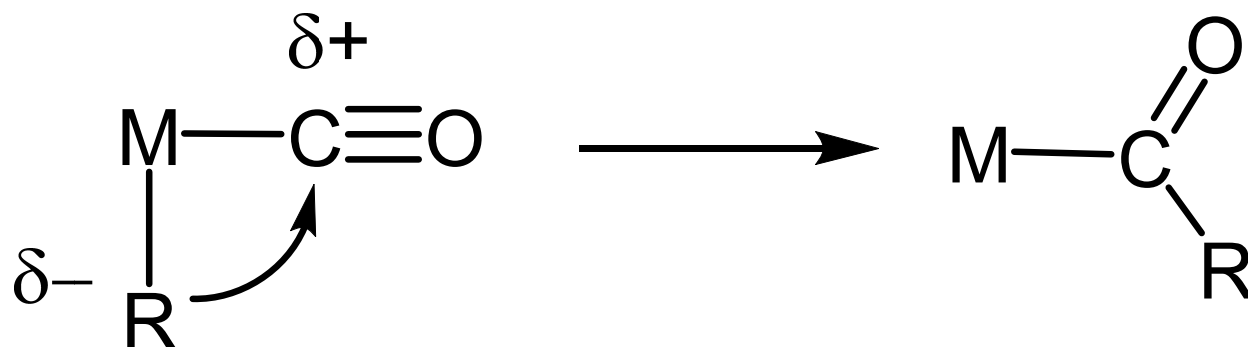
1. There is **no change** in the formal oxidation state of the metal unless AB is an alkylidene or alkylidyne.
2. The groups undergoing migratory insertion must be **cis to one another**. In complexes where the **cis coordination sites are blocked** by strongly coordinated ligands insertion or elimination processes are **not possible**.
3. An open coordination site is created during migratory insertion. Therefore, for the reverse reaction (elimination) to occur, an open coordination site must be generated by ligand dissociation.
4. In cases where C is a chiral carbon center, the reaction usually occurs with **retention of configuration**.
5. Cases where C migrates to AB followed by coordination of L in place of C, and where AB migrates to C followed by coordination of L in place of AB are both known.

B. CO Insertion reactions

There are many examples of CO insertions. *e.g.*

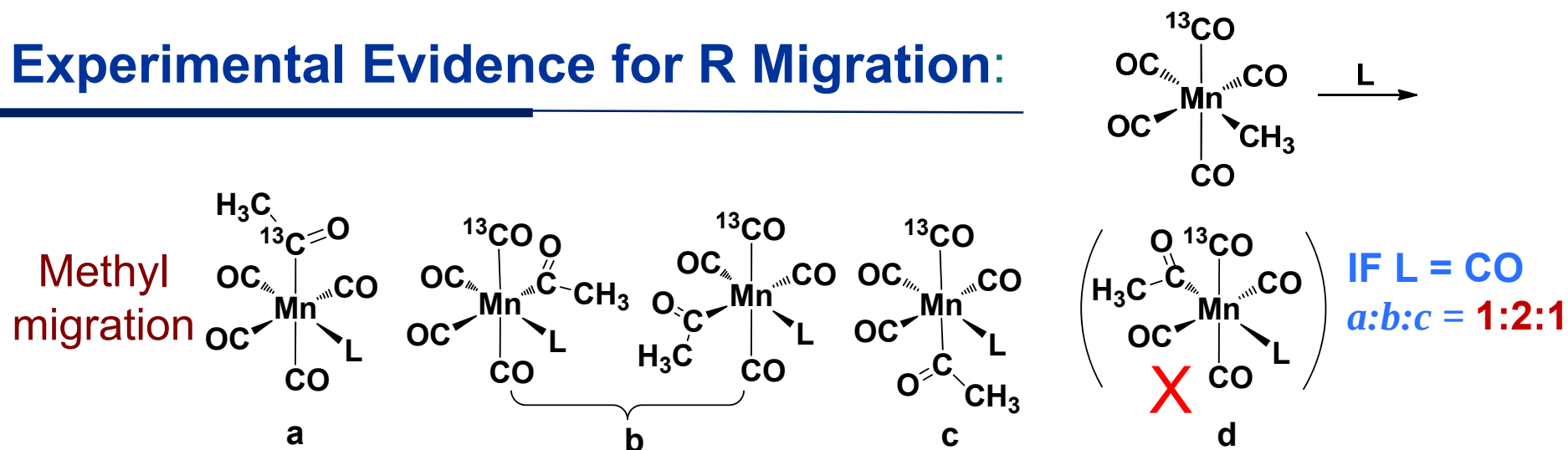


1) Main features of CO insertion reactions

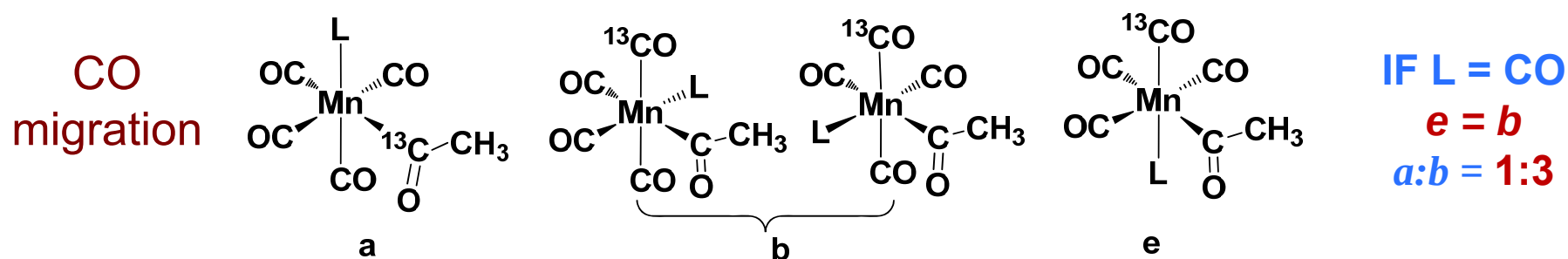


- a). CO and R **must be *cis***.
- b). It **can be described as R migration**
- c). The **configuration of R is unchanged**.
- d). The feasibility and rate depend on M-R bond strength.

Experimental Evidence for R Migration:



Methyl migration to CO followed by ligand coordination at the position vacated by Me cannot give **d**.

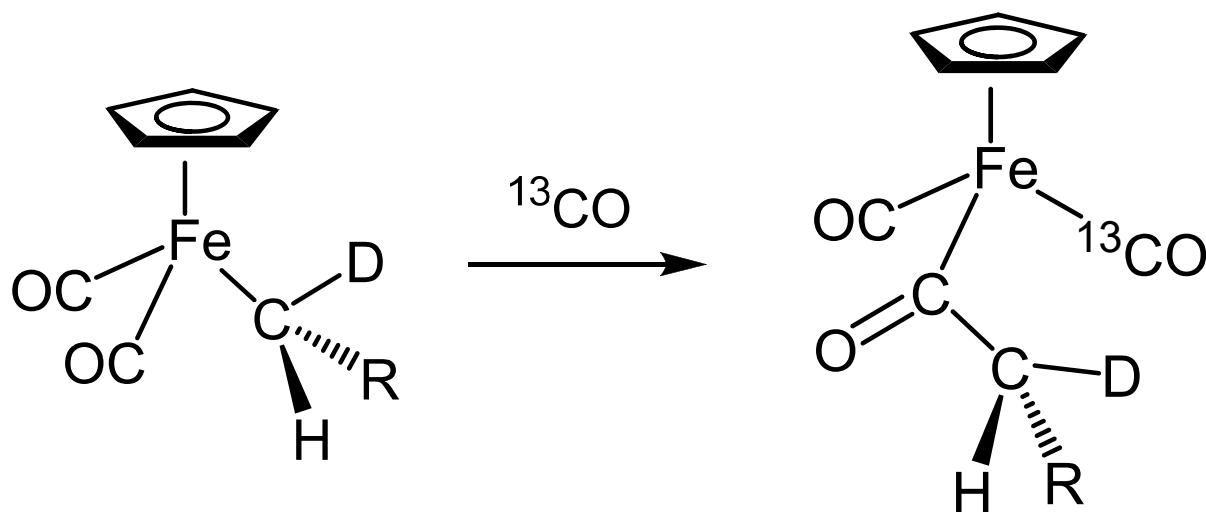
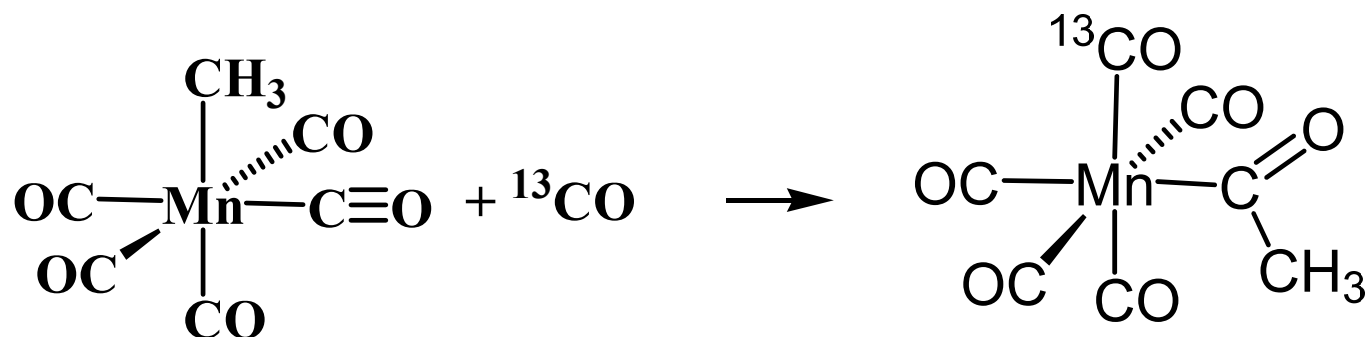


IF the incoming ligand is CO, a 1:2:1 ratio of $a:b:c$ would be expected for *methyl migration*, while in the case of *CO migration*, **e** is also the isomer of **b**, thus a 1:3 ratio of $a:b$ would be expected.

The experimental results (1:2:1) are consistent with methyl migration.

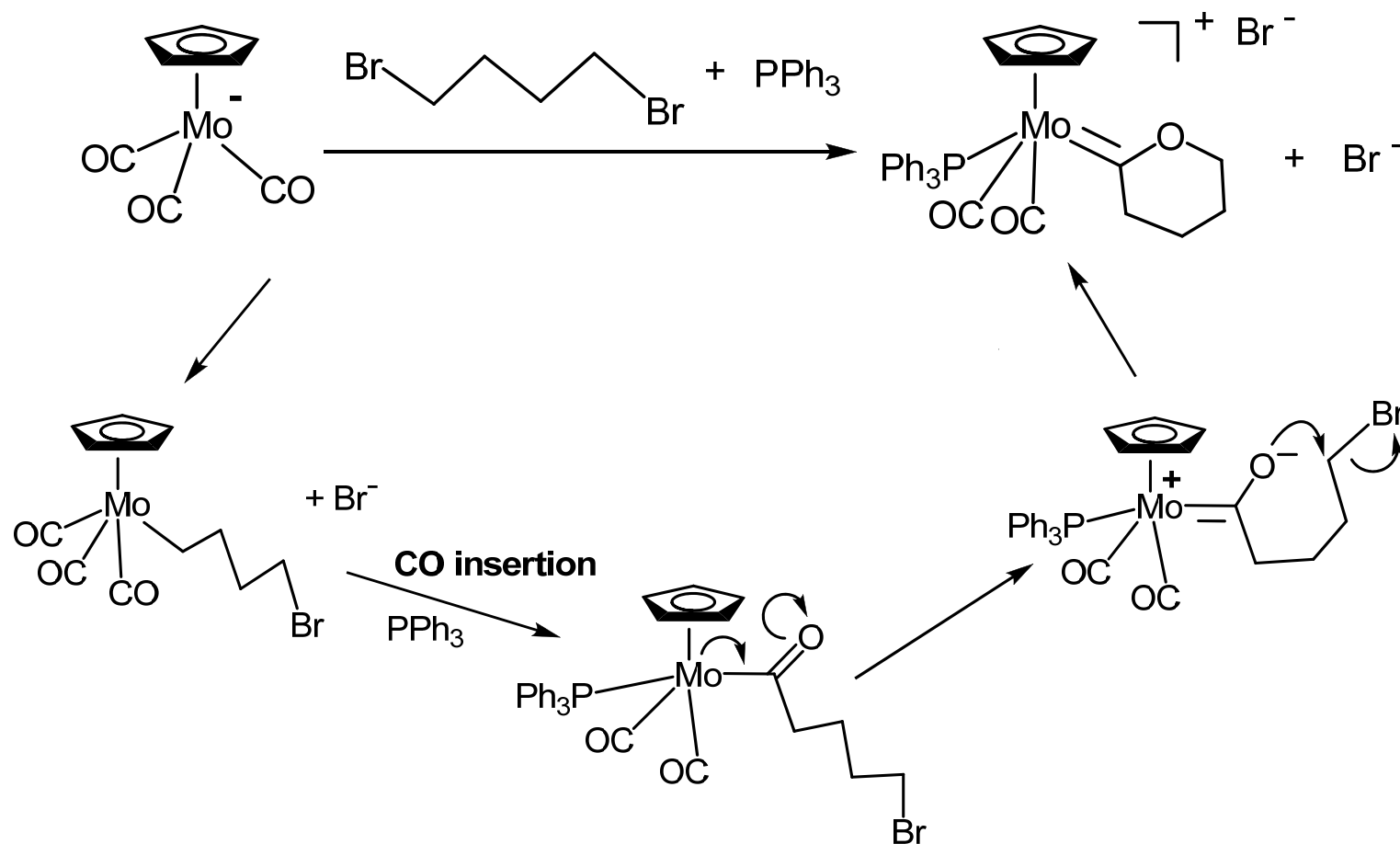
Practice Problems

Exercise 1. Suggest product for the following reactions

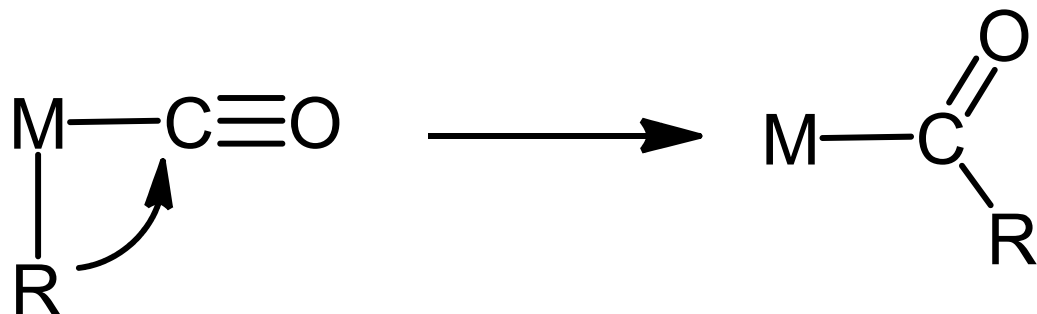


Practice Problems

Exercise 2. propose a mechanism for the reaction:



2). Thermodynamics of CO insertions

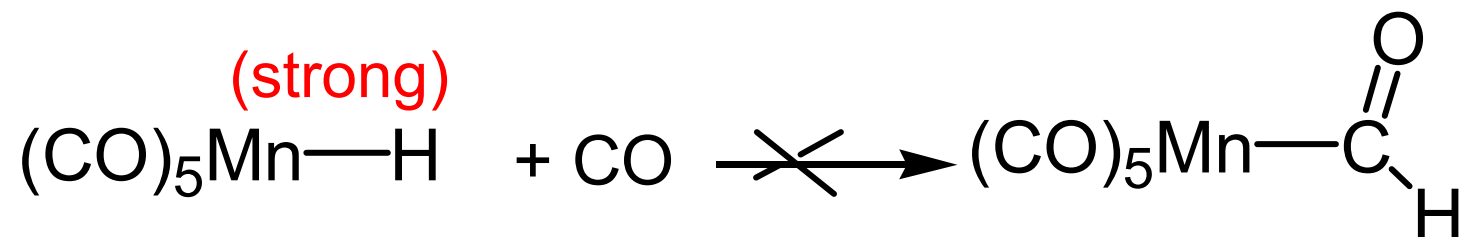
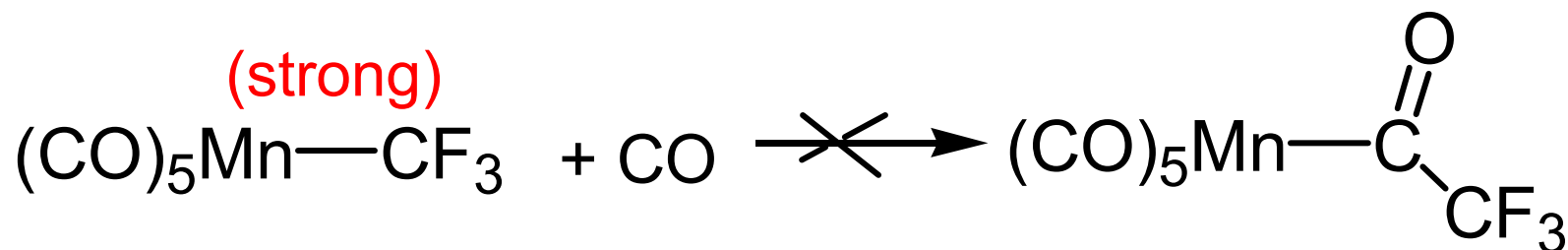
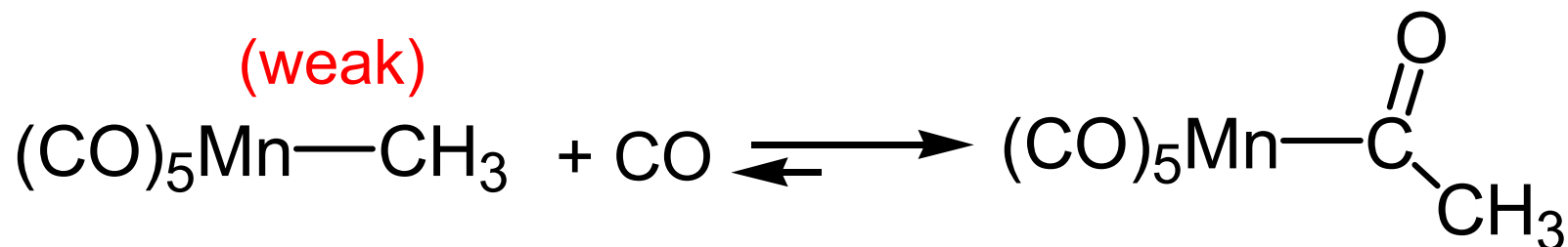


Bond strength of M-R is a determining factor.

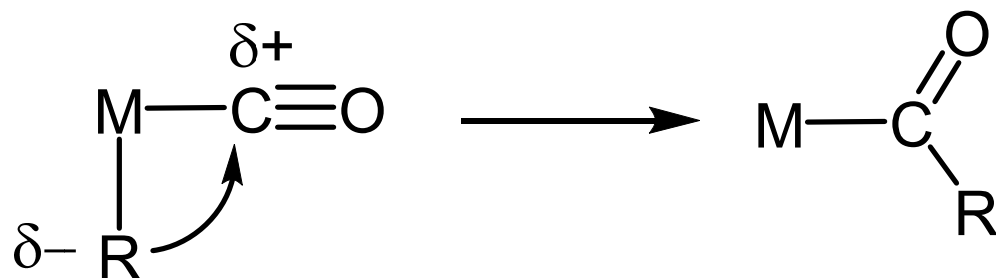
The weaker the M-R bond, the easier to undergo a insertion reaction and the more favorable thermodynamically.

Practice Problems

Explain the following observations.



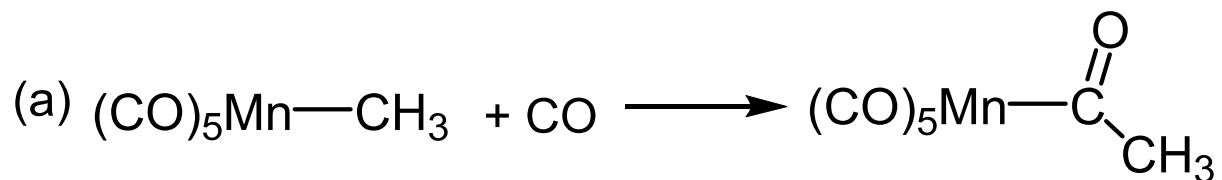
3). Factors influencing the insertion reaction rates



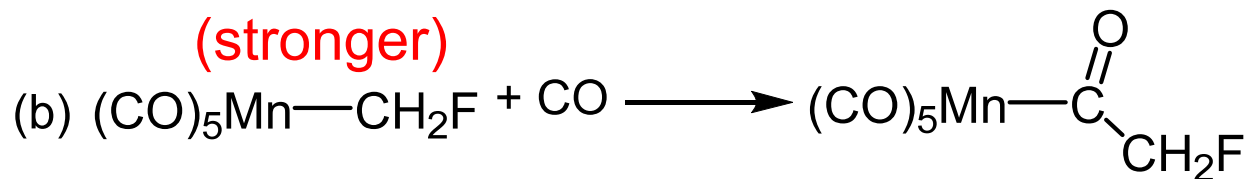
General expectation: rate will increase when

- a) More positive charge on C atom of CO;
- b) Weak M-R bond strength.

Practice Problem: Which of the following reaction is faster?

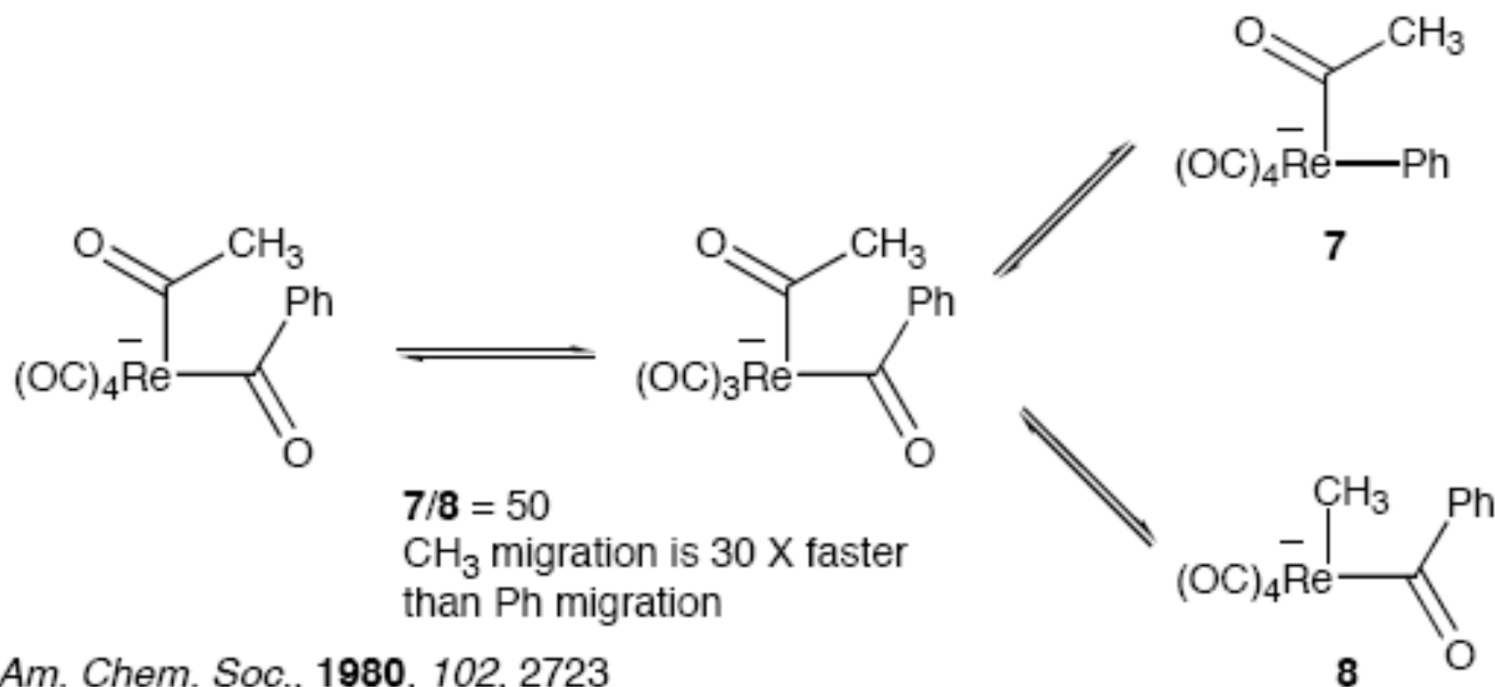


Answer: (a)

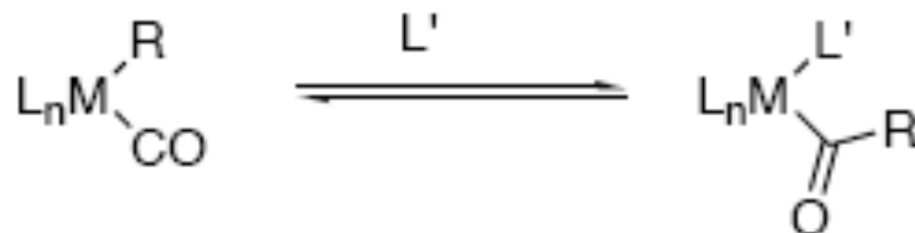


(stronger)

The identity of R has a significant effect on the aptitude to undergo migratory insertion to form an metal acyl.



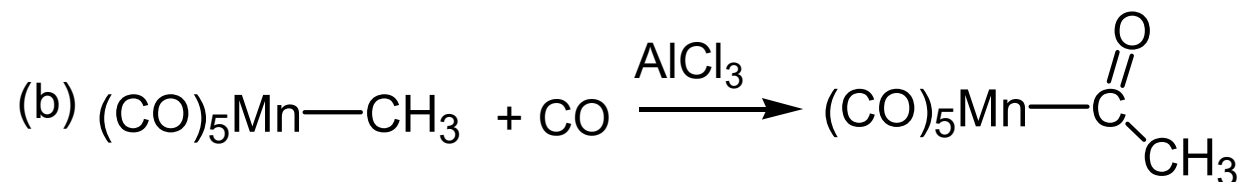
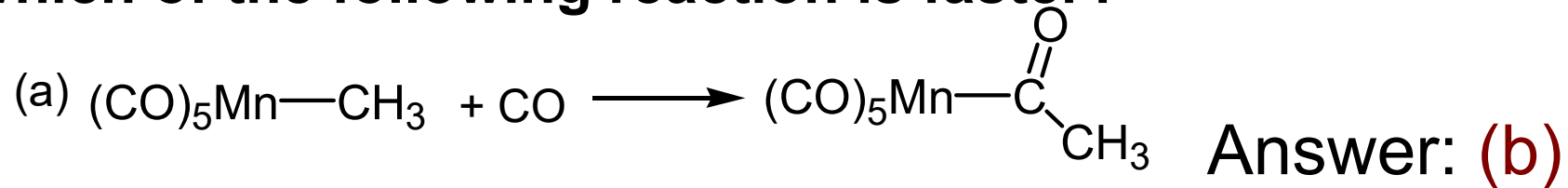
Cautious: The identity of R has a significant effect on the aptitude to undergo migratory insertion to form an metal acyl. **However, Migratory insertion ability may not be correlated to M-R bond strength in some cases.**



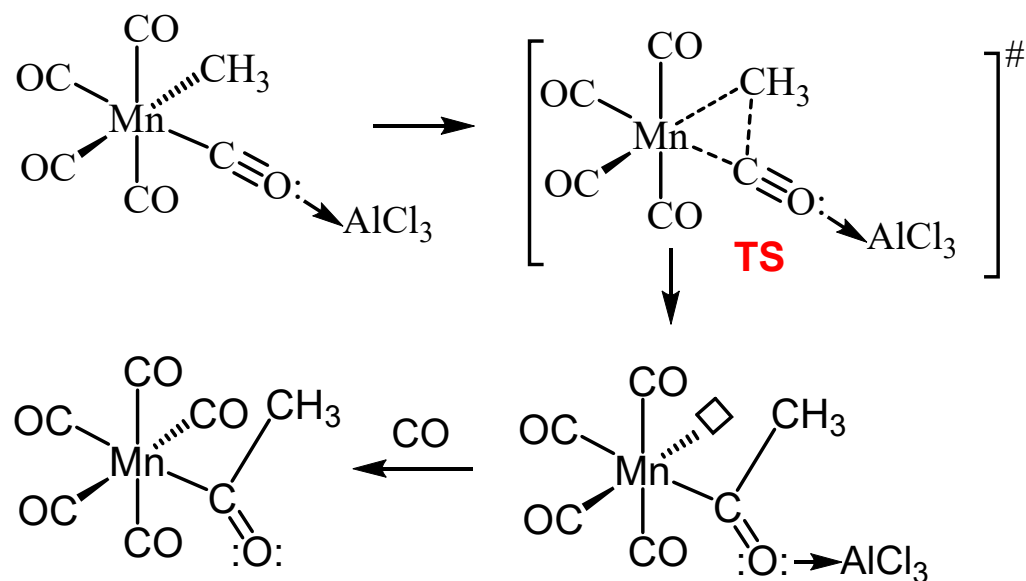
R	M-L BDE	Insertion/Elimination
PhCH ₂	29	Slower insertion than CH ₃
CH ₃ CO-	39	Elimination
CH ₃	44	Fast insertion
CF ₃	49	Elimination
Ph	49	Slower insertion than CH ₃ , but the product is more stable
H	59	Elimination

Practice Problems

Which of the following reaction is faster?



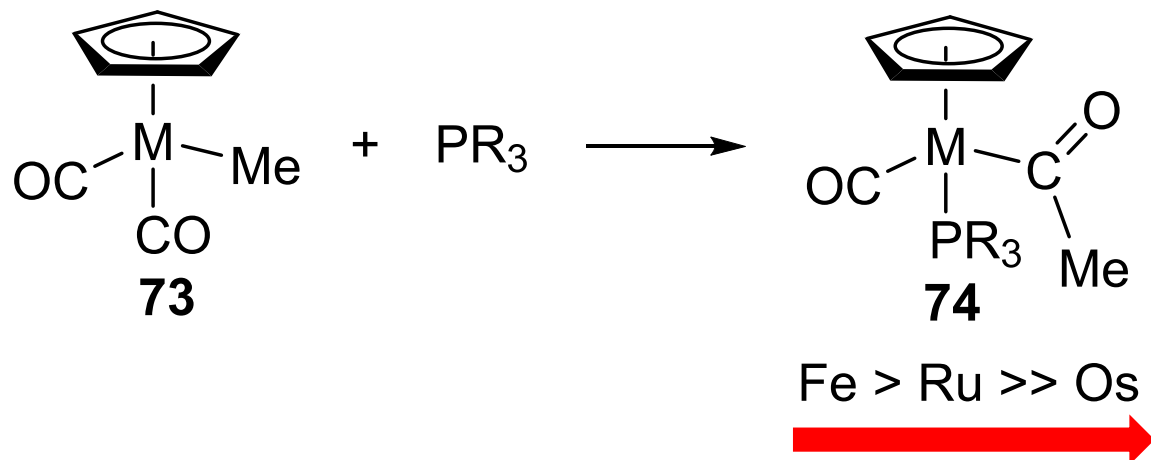
Explanation:



Similarly, cations such as Na⁺, Mg²⁺ can also enhance the insertion reaction rate.

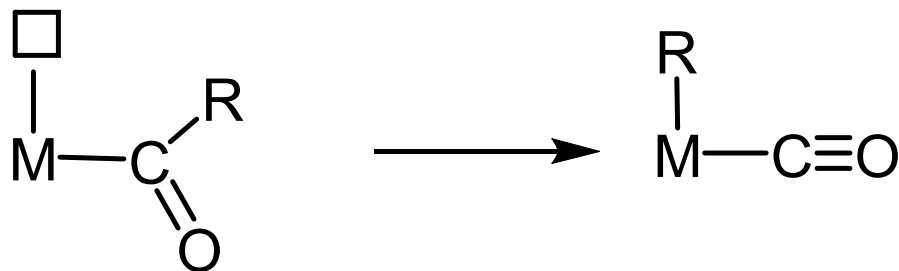
Practice Problems

Explain the following observations:



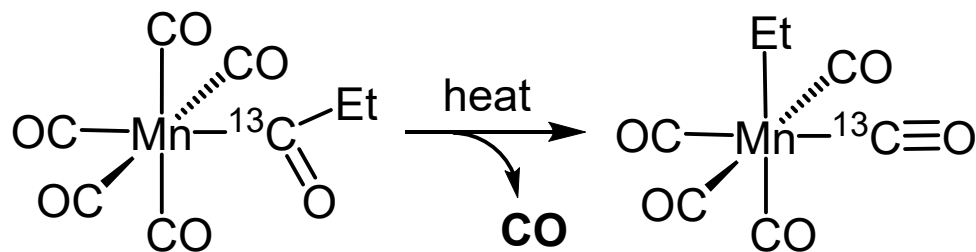
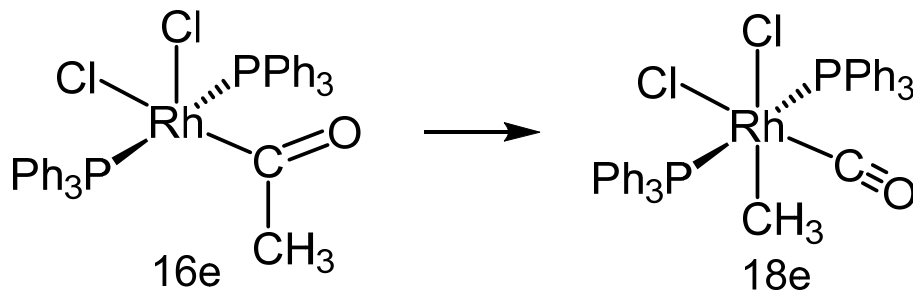
The transition metal-alkyl bond strength increases when going down a group.

4). Decarbonylation reactions (CO de-insertion)



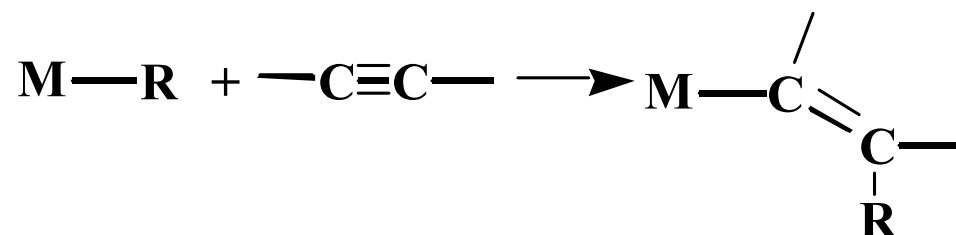
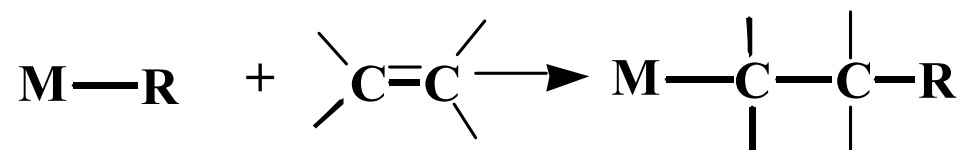
- The complexes must be unsaturated coordinatively.
- The CO and R are *cis* to each other in the products.
- The rate is dependent on R.

Examples



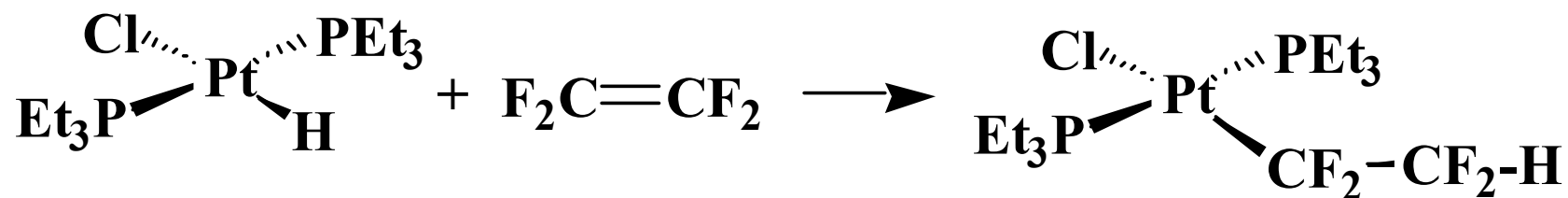
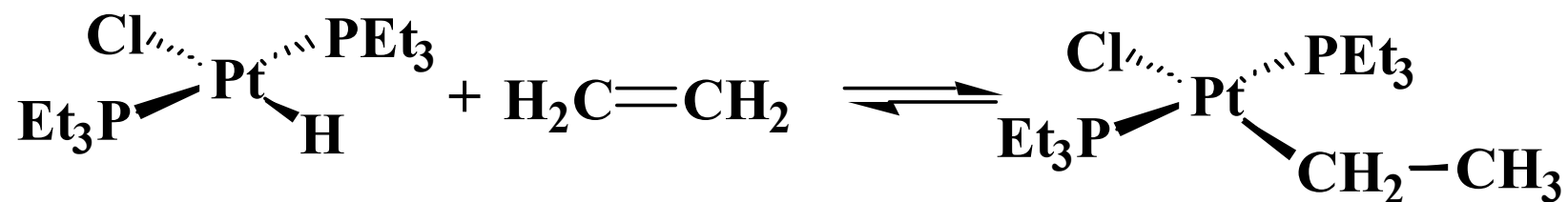
(One CO ligand leaves from metal center firstly.)

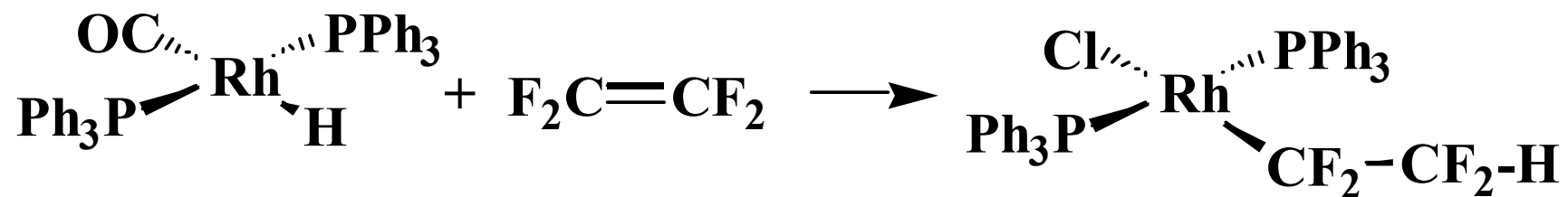
C. Insertion reactions involving alkenes and alkynes



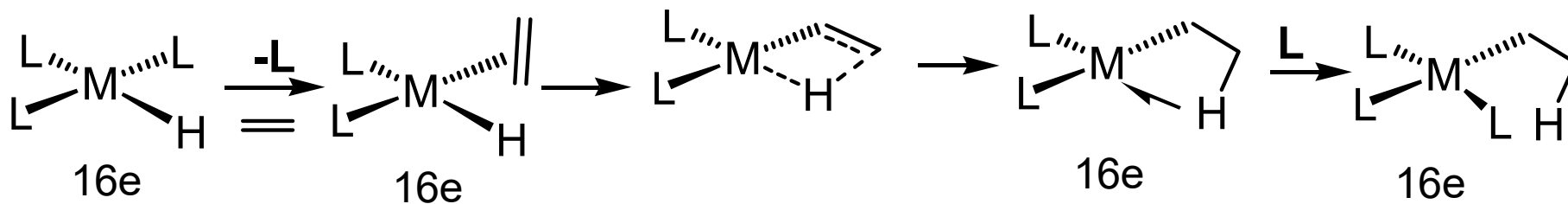
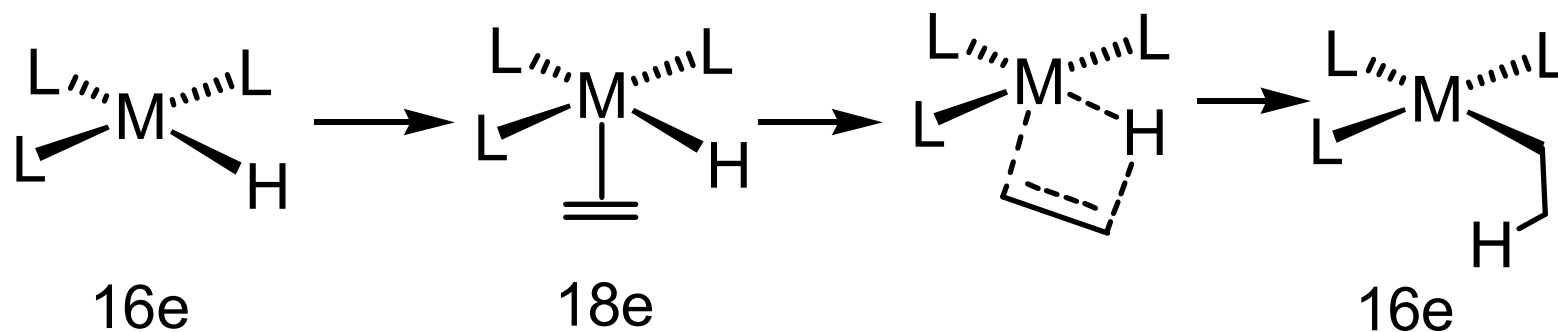
Very important in
catalysis

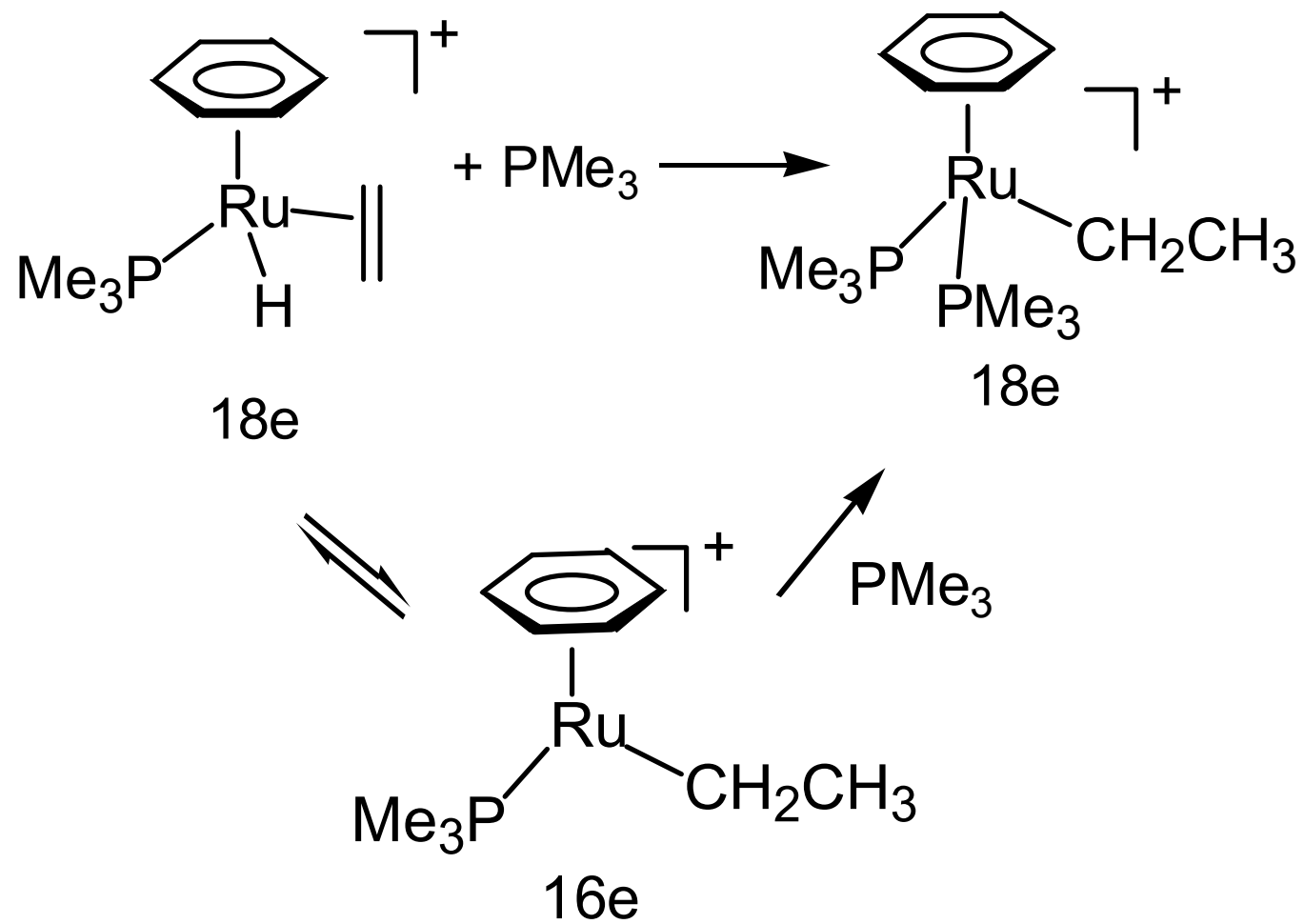
1). Examples of alkene and alkyne insertion reactions





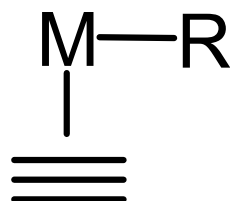
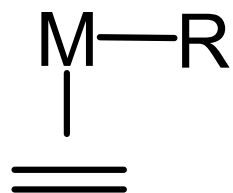
Mechanistically,



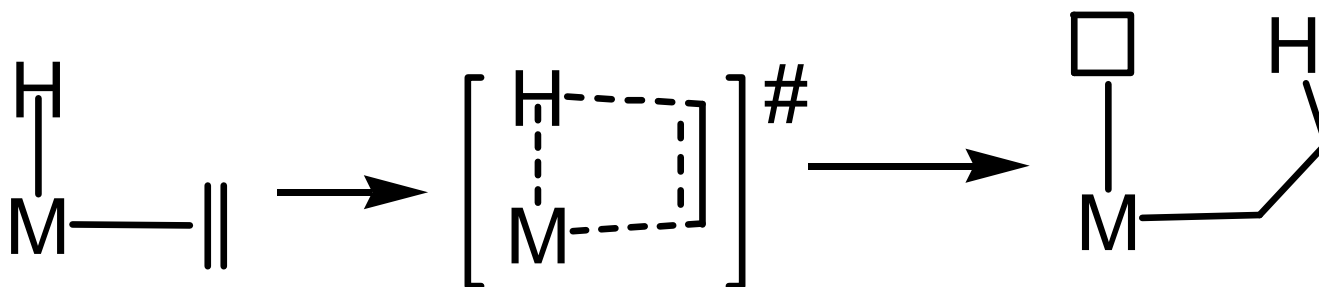


2). Some features of alkene and alkyne insertion reactions

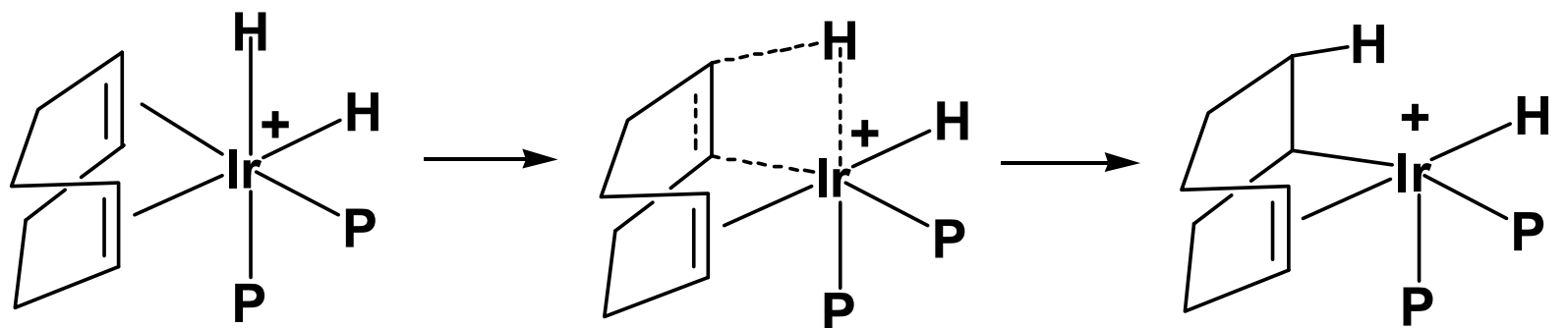
a). Need an intermediate in which the olefin or alkyne is *cis* to R.

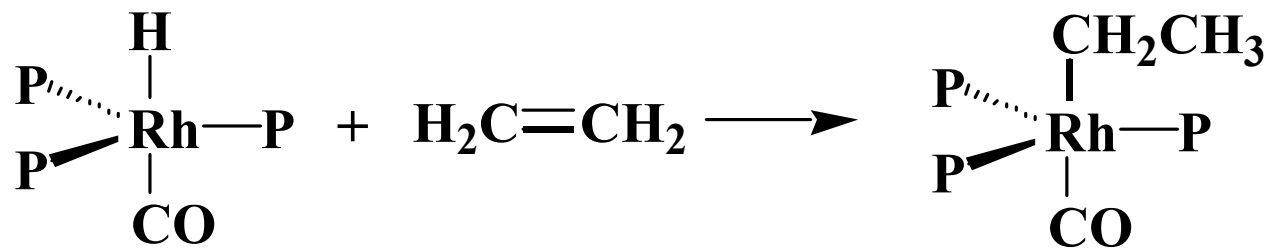


b). The transition state has a coplanar geometry.



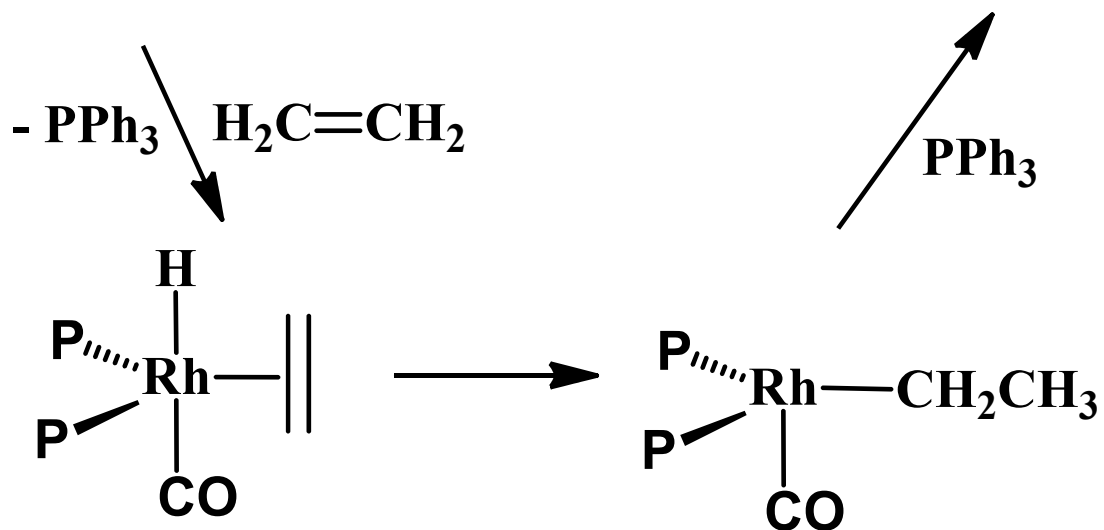
Example:



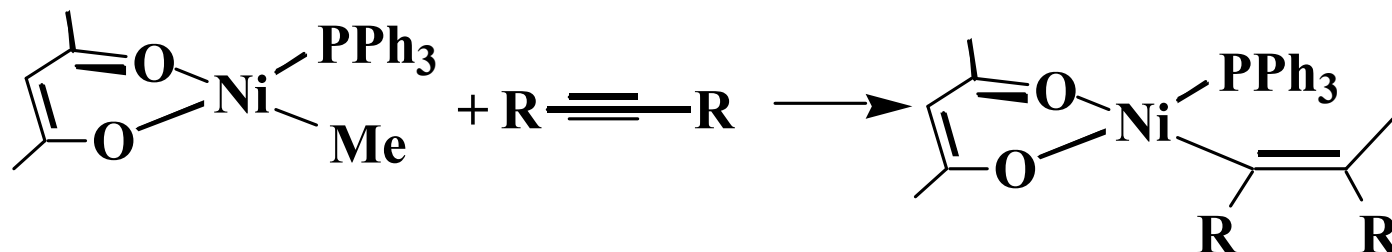
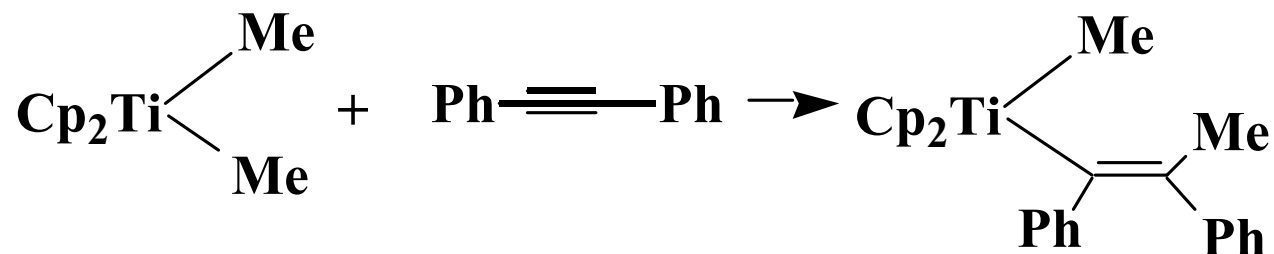
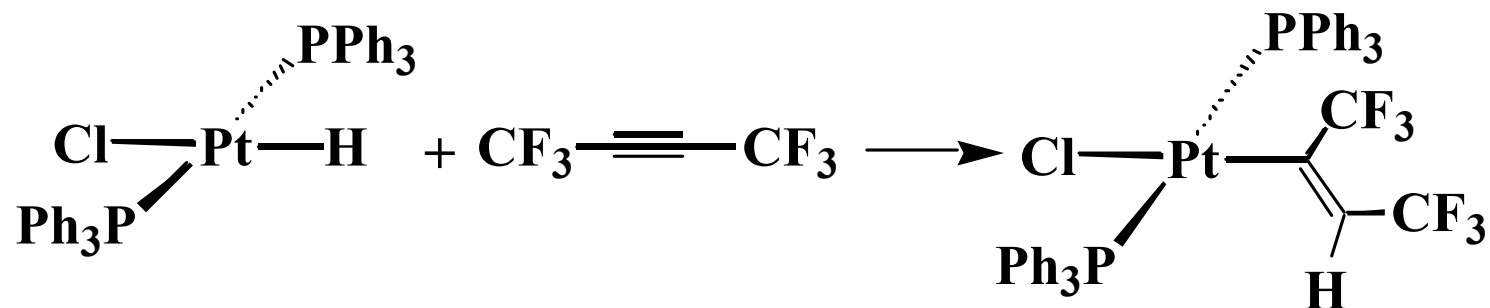


$\text{P} = \text{PPh}_3, 18\text{e}$

Mechanism?

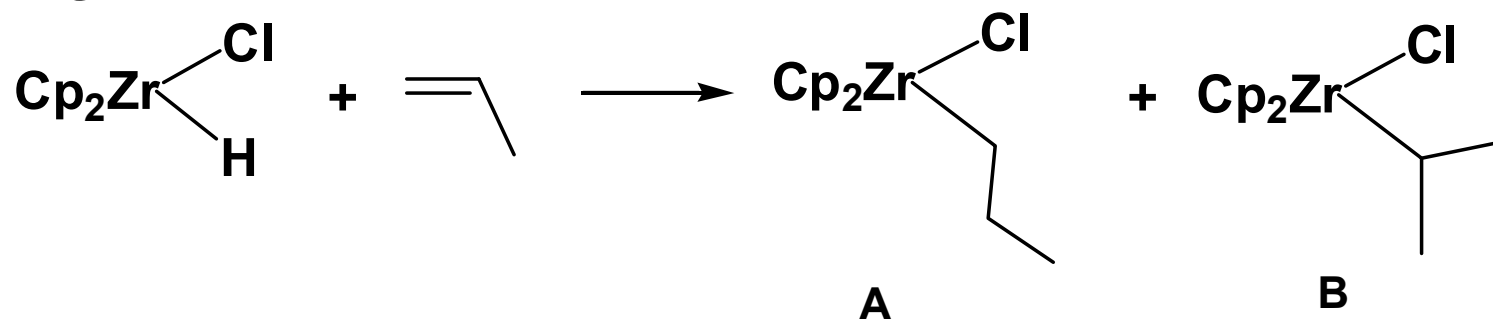


c). The stereochemistry is *syn* for the two substituents of $C\equiv C$. *e.g.*

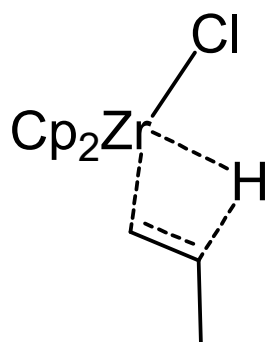


d). For substituted olefins, there are two possible structure.

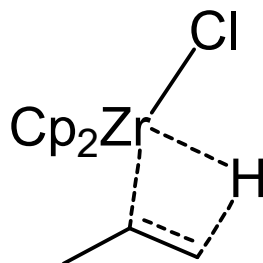
e.g.



Which one is major ?



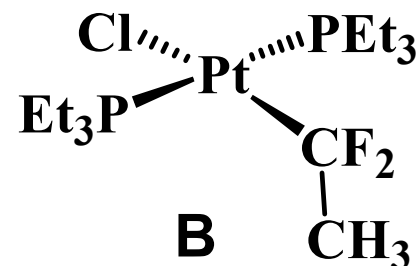
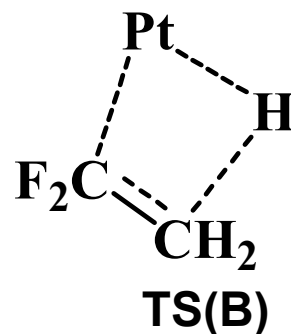
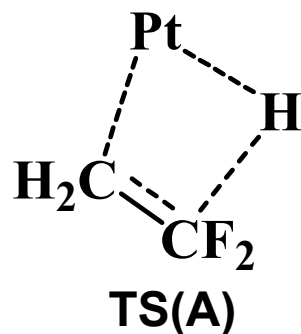
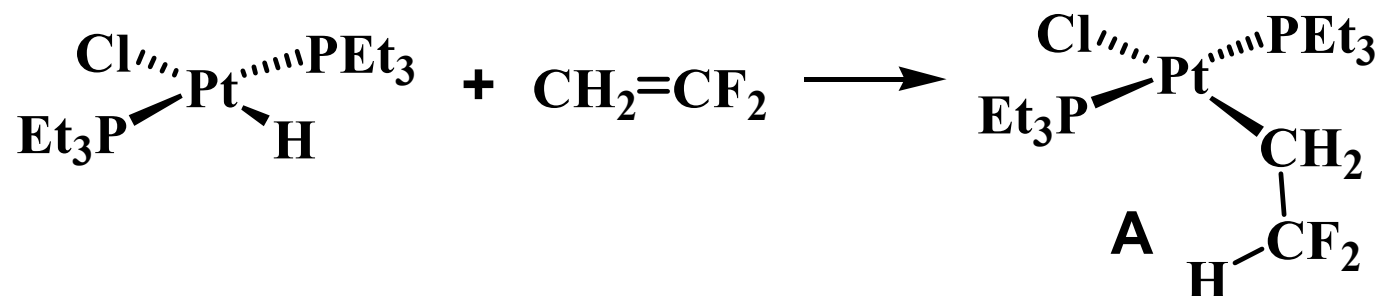
TS(A)



TS(B)

Answer :

A (less steric hindrance)



Which one is major ?

Answer : **B**

(electronic reason, F is electron withdrawing group, CH₂ is more positive than CF₂)

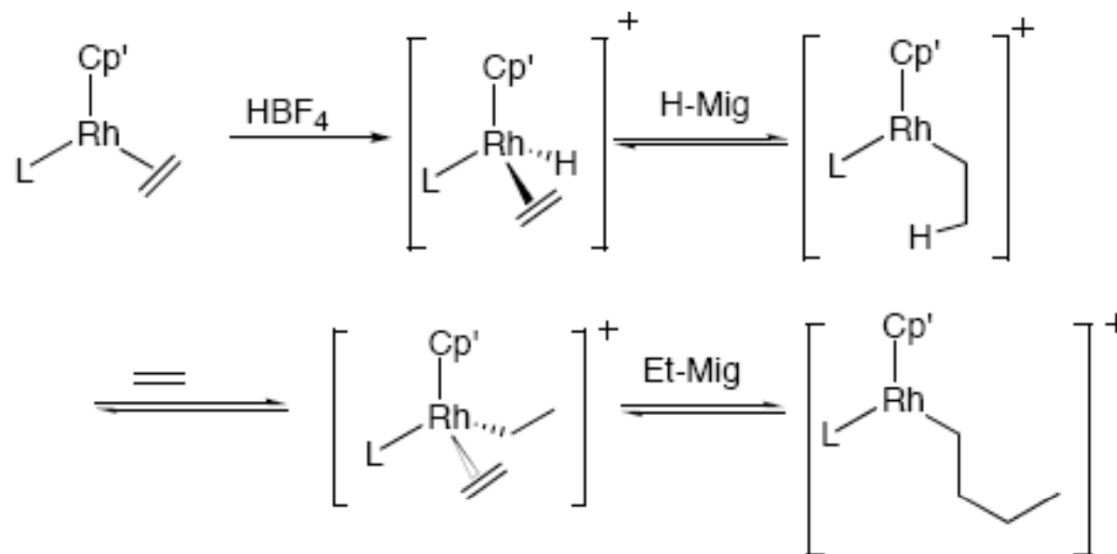
Insertion into M-H versus M-R

Migratory insertion into M-H bonds occurs at much higher rates than insertion into M-R bonds.

Alkene insertion into M-R is thermodynamically favored over M-H insertion.

Therefore, the preference for M-H insertion must be due to kinetic factors, not thermodynamic factors.

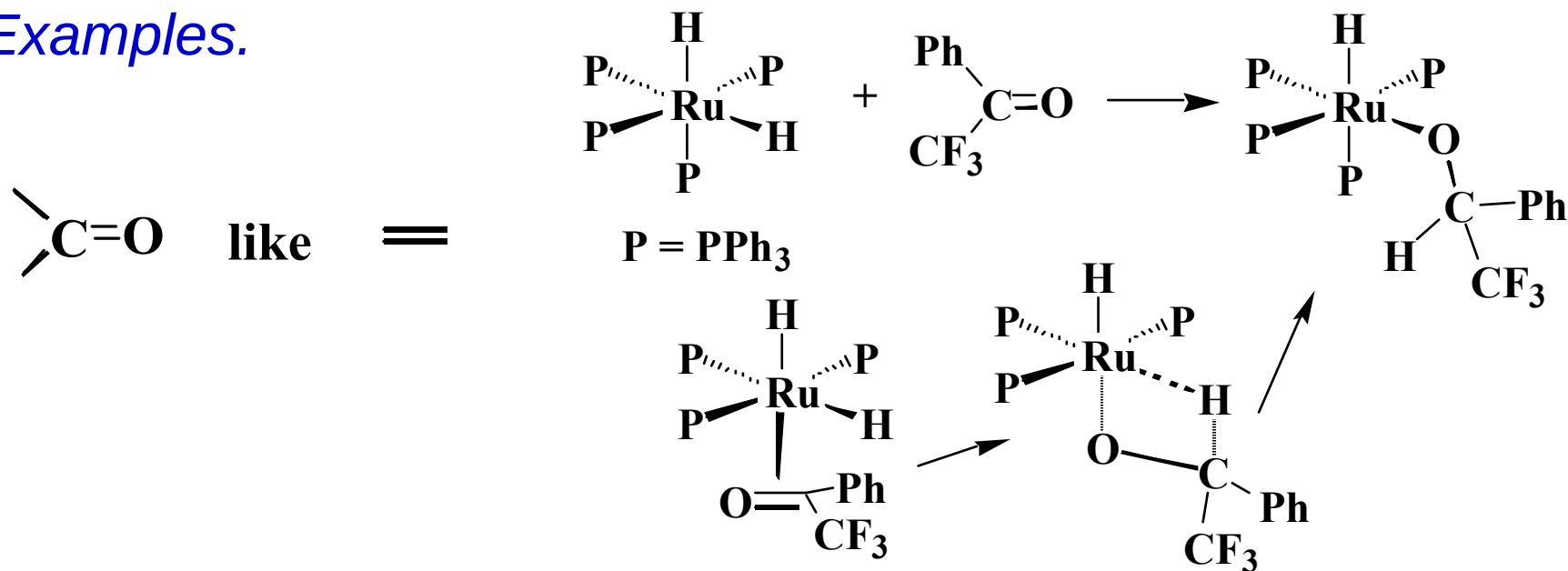
One example of the difference in activation energies was reported by Brookhart (*J. Am. Chem. Soc.*, **1992**, *114*, 10394)



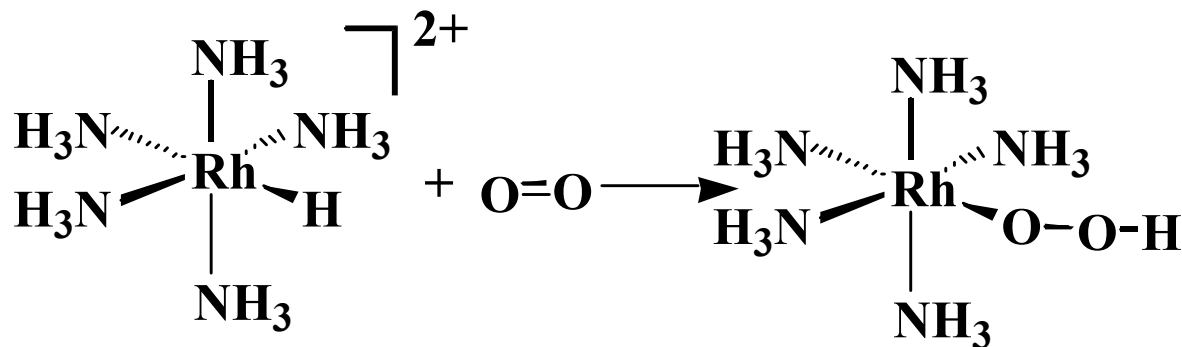
Cp'	L	$\Delta G^\ddagger$ H-Mig (kcal/mol)	$\Delta G^\ddagger$ Et-Mig (kcal/mol)
Cp*	P(OMe) ₃	12.2	22.4
Cp*	PMe ₃	12.1	23.4
Cp	P(OMe) ₃	15.0	23.4
Cp	PMe ₃	15	24.7

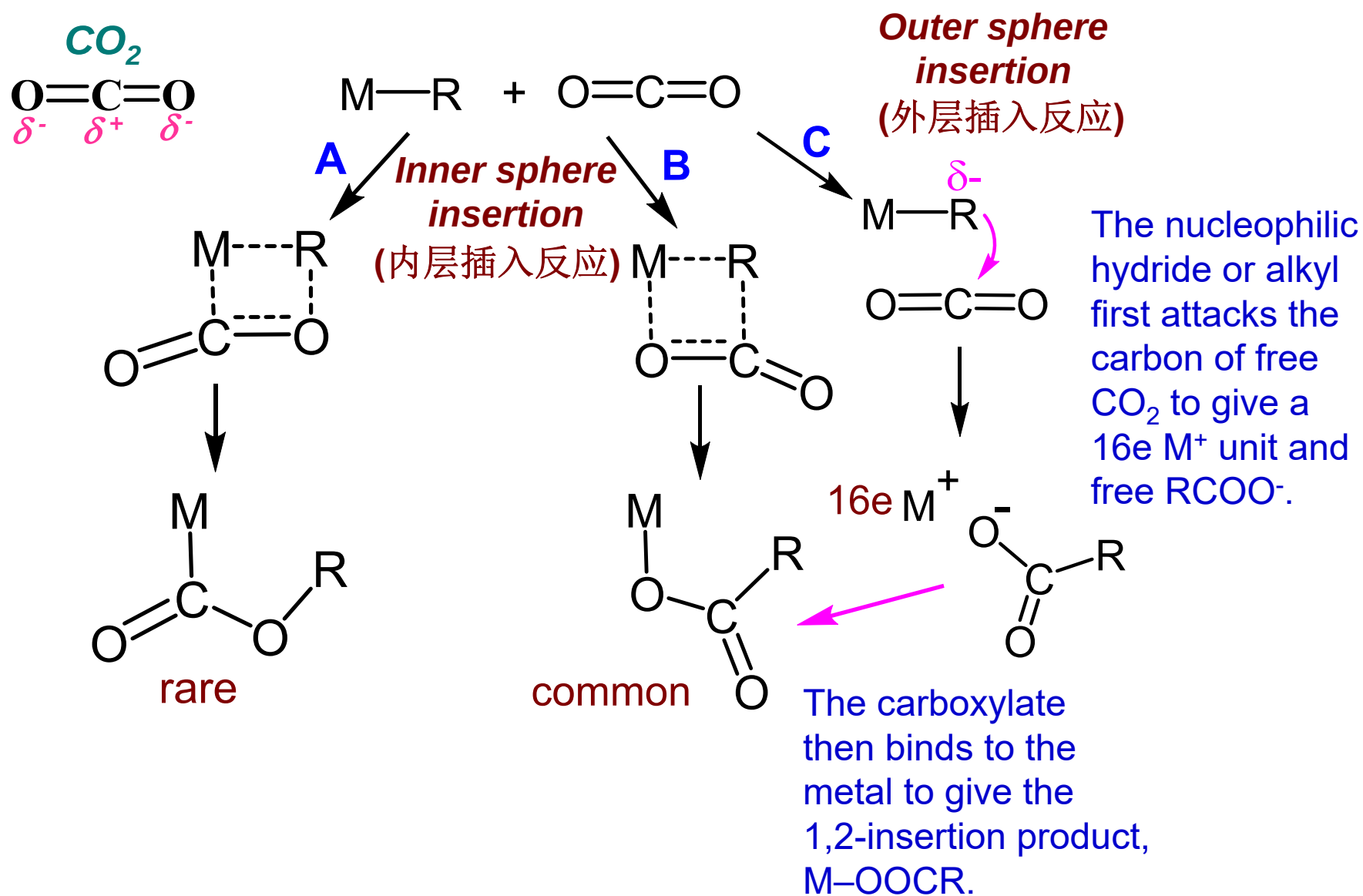
e). Related insertion reactions.

Examples.



$\text{O}=\text{O} \quad \text{like} \quad =$



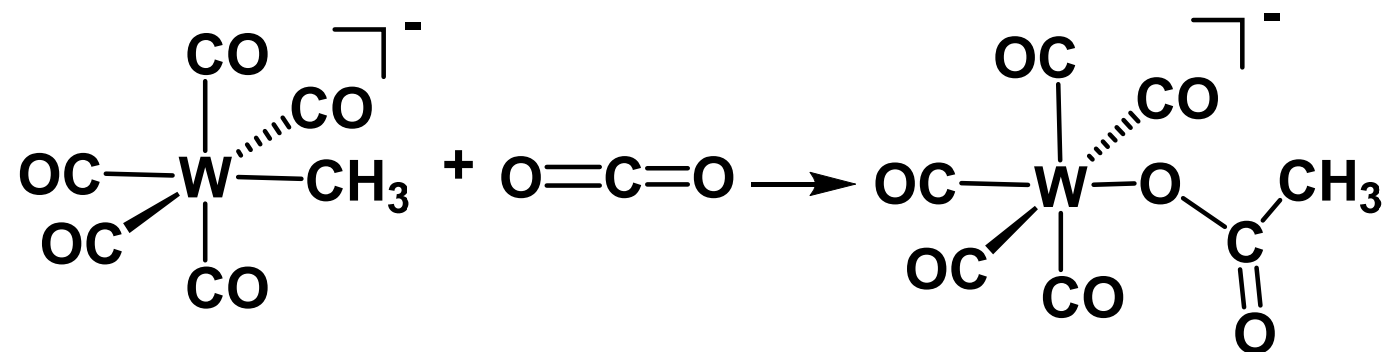


Outer Sphere Insertions (外层插入反应)

In some cases, the A=B bond does not need to coordinate to the metal prior to insertion and can undergo the reaction with an 18e complex.

The weakly binding ligand, CO₂, can insert into an M-H or M-R bond in this way.

Example.



D. α , β , γ and δ elimination reactions.

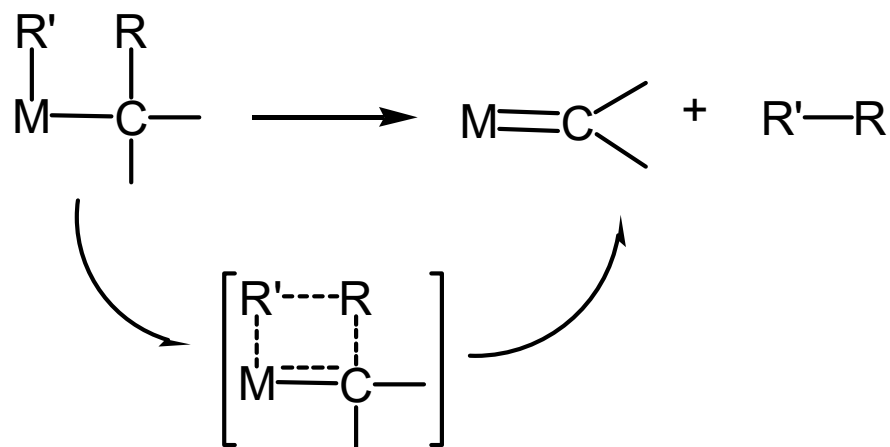
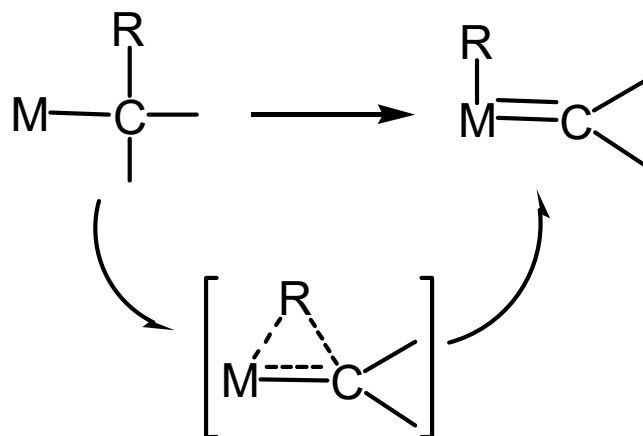
1). α elimination

2). β elimination

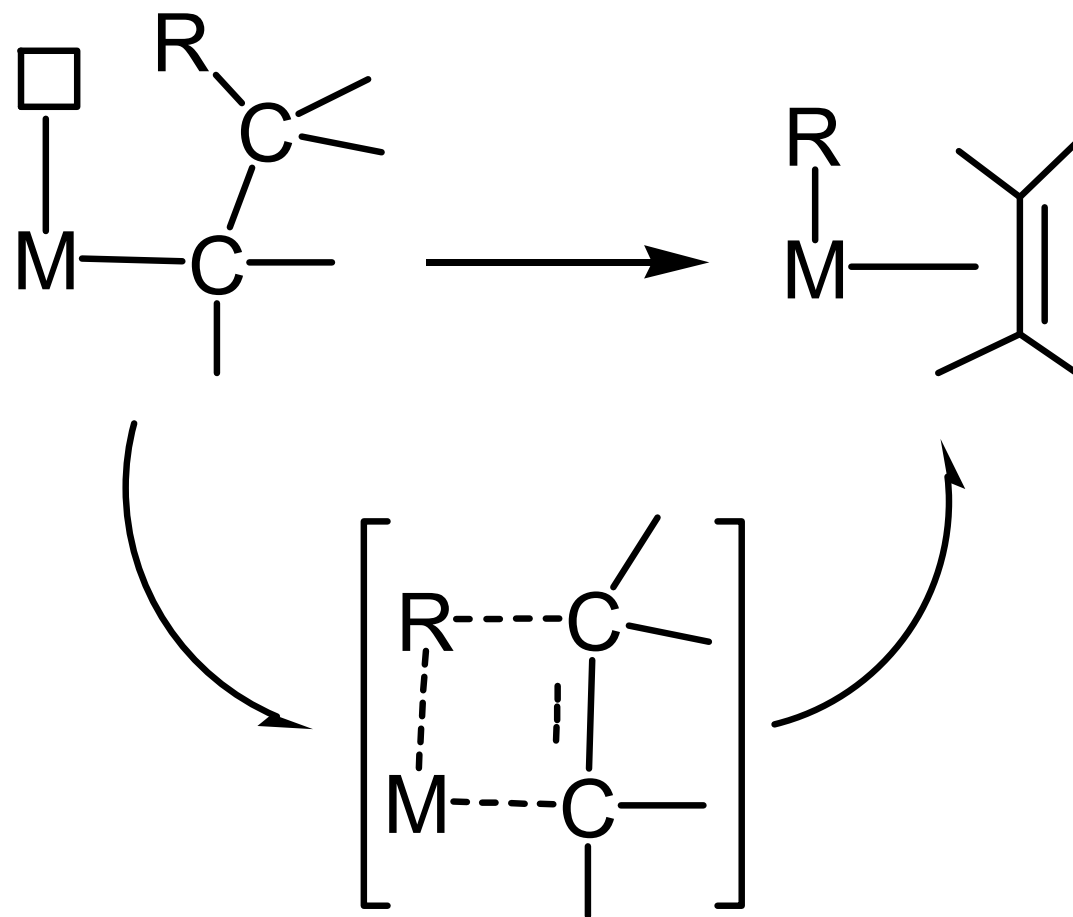
3). γ elimination

4). δ elimination

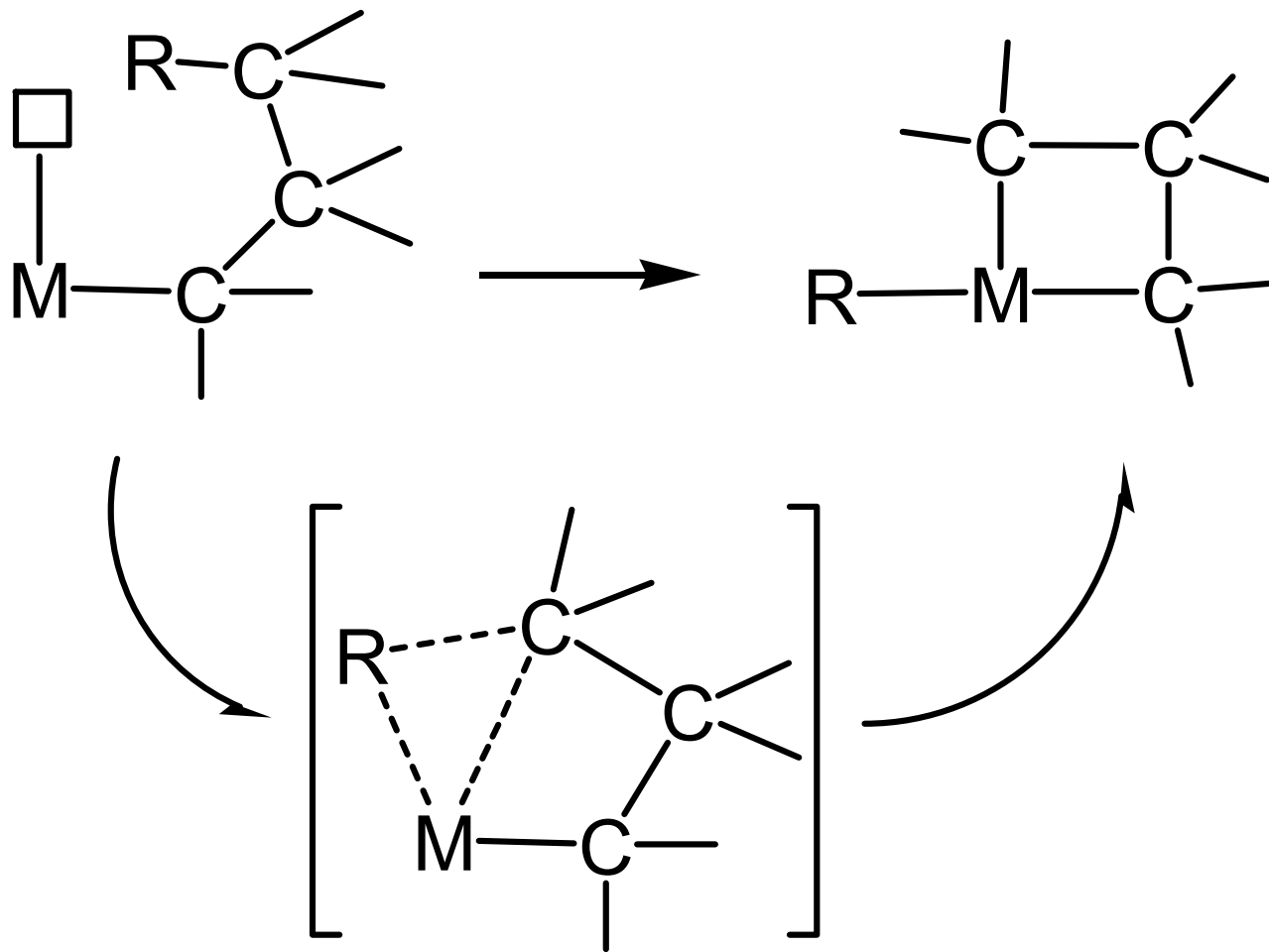
1). α -elimination



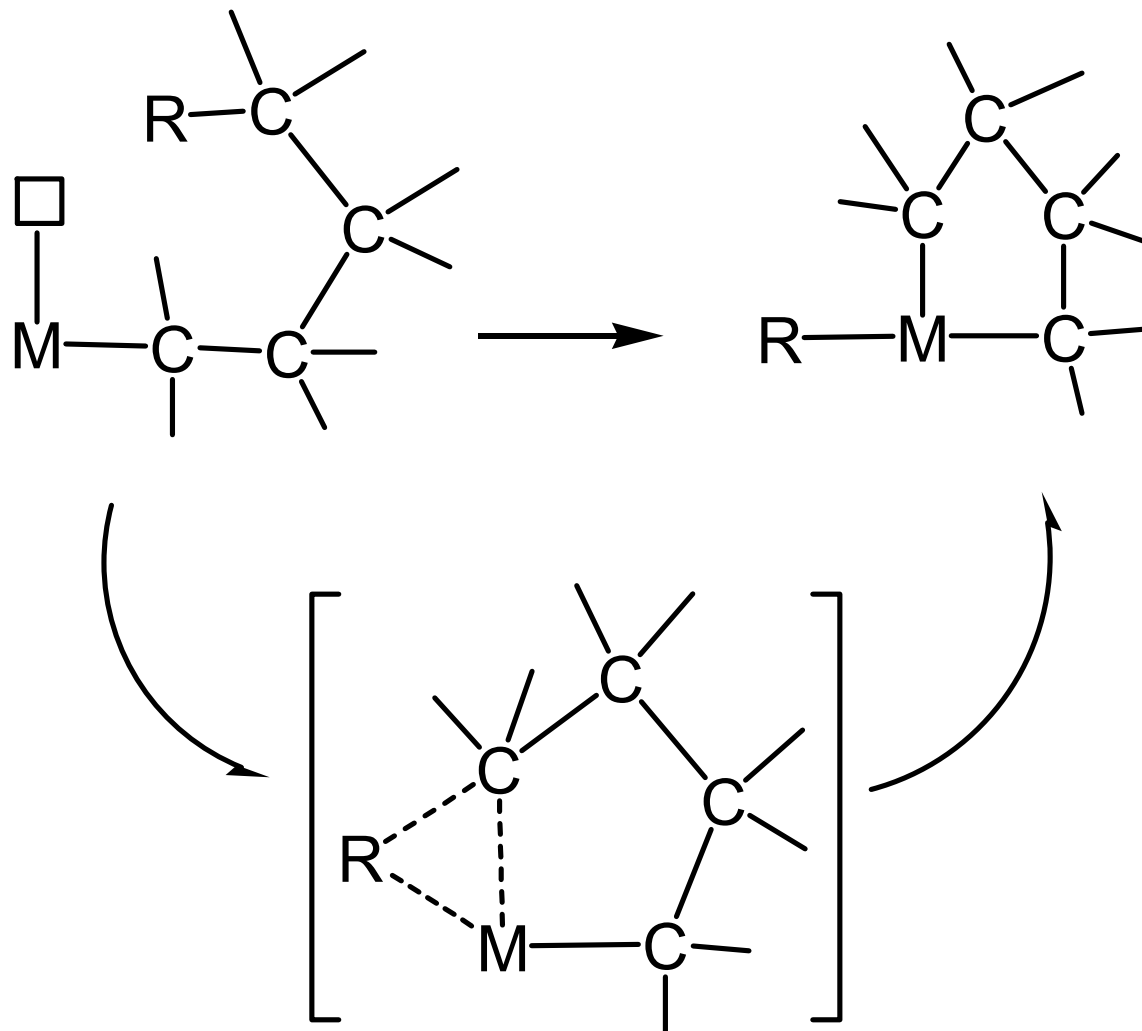
2). β -elimination



3). γ -elimination

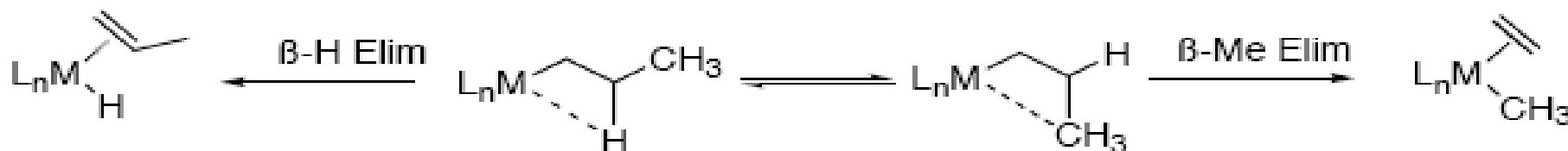


4). δ -elimination

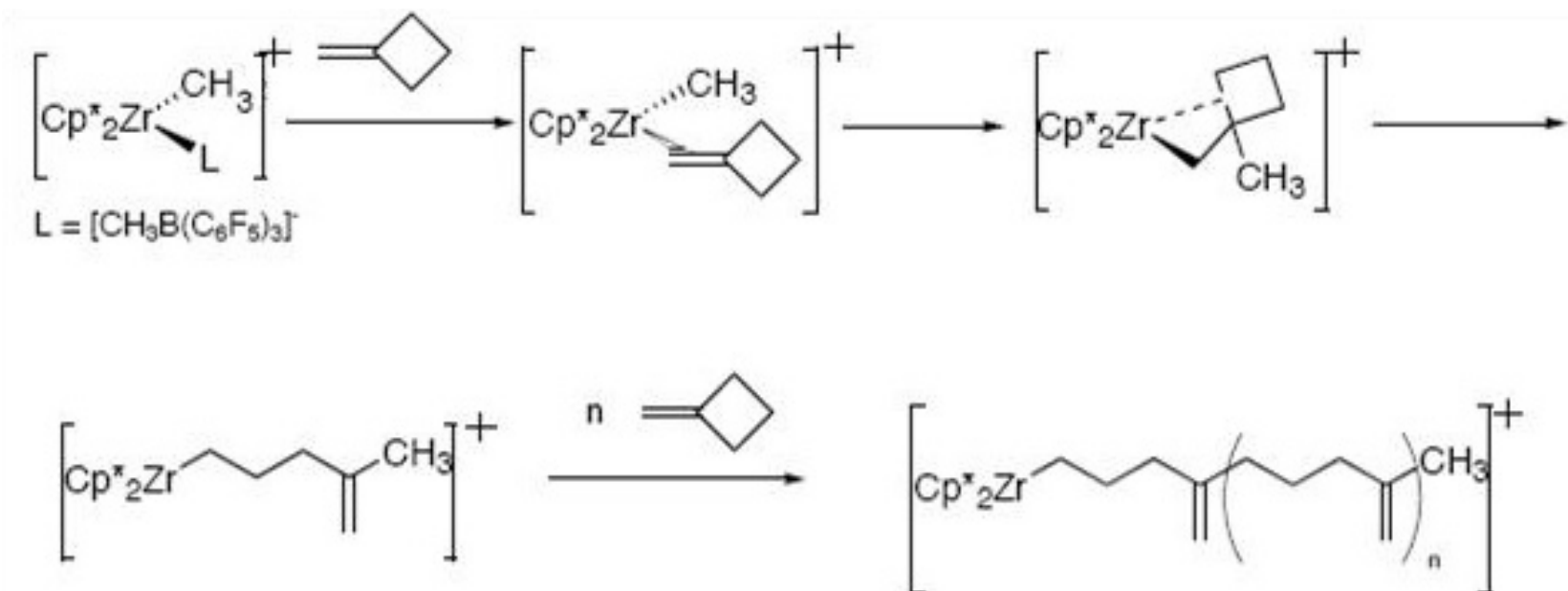


β-Carbon Elimination

In general, there is preference for β-H elimination over β-alkyl elimination.

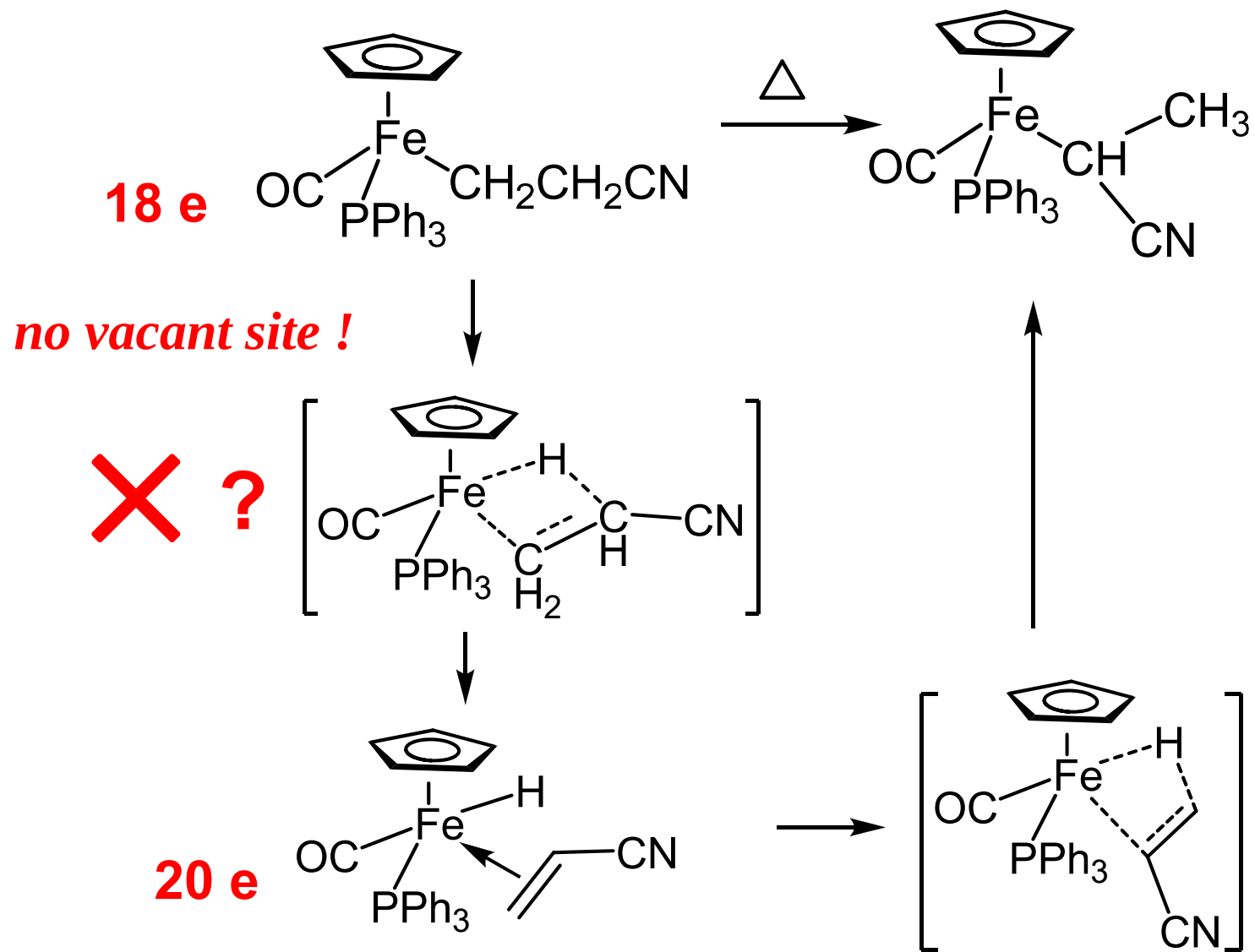


Typically β-alkyl elimination is only observed where there is a **large** amount of strain in the C-C bond, or where there is no β-H to undergo elimination. β-Alkyl elimination is **more common** with highly electrophilic d^0 metal complexes.

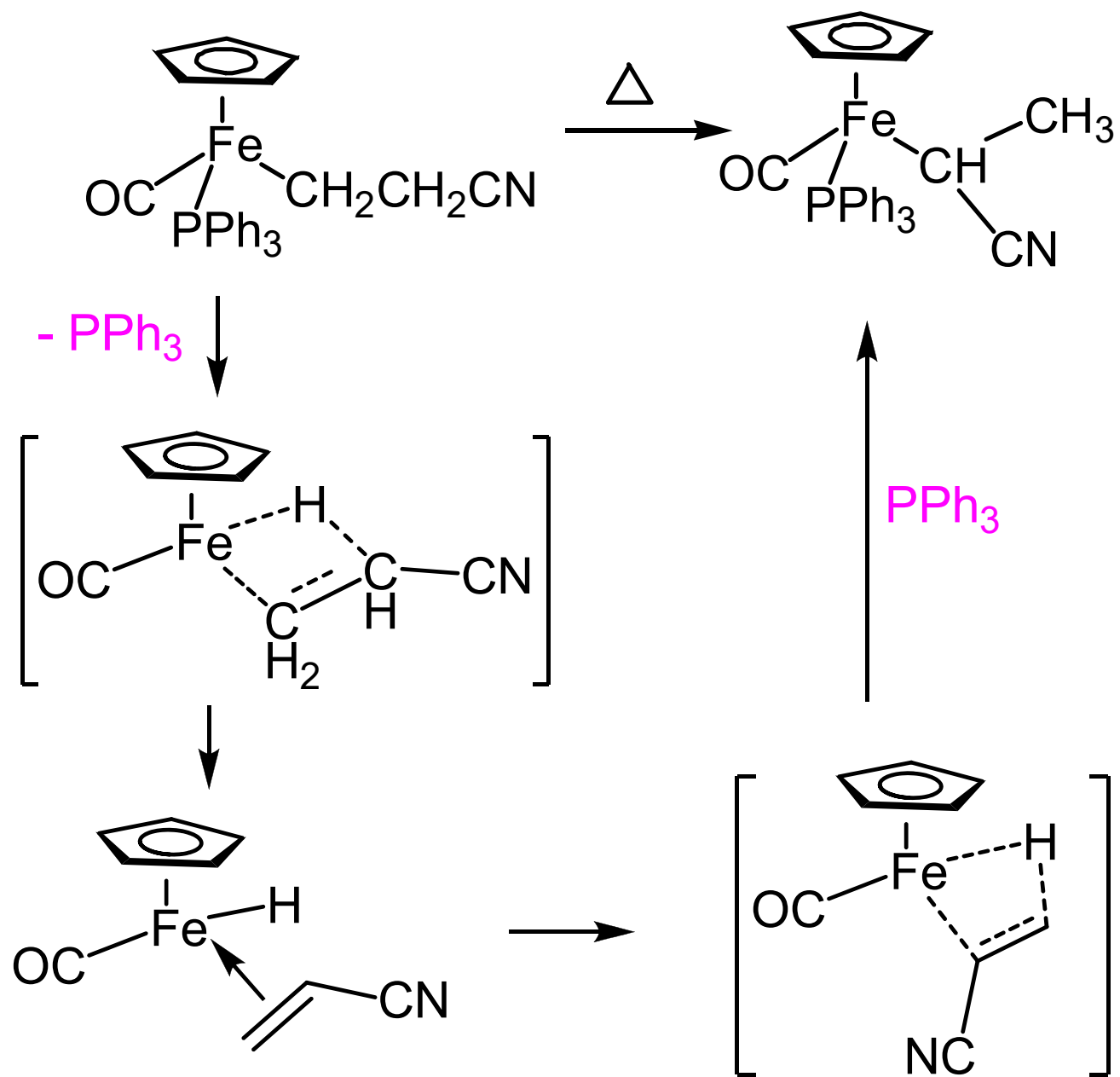


Practice Problems

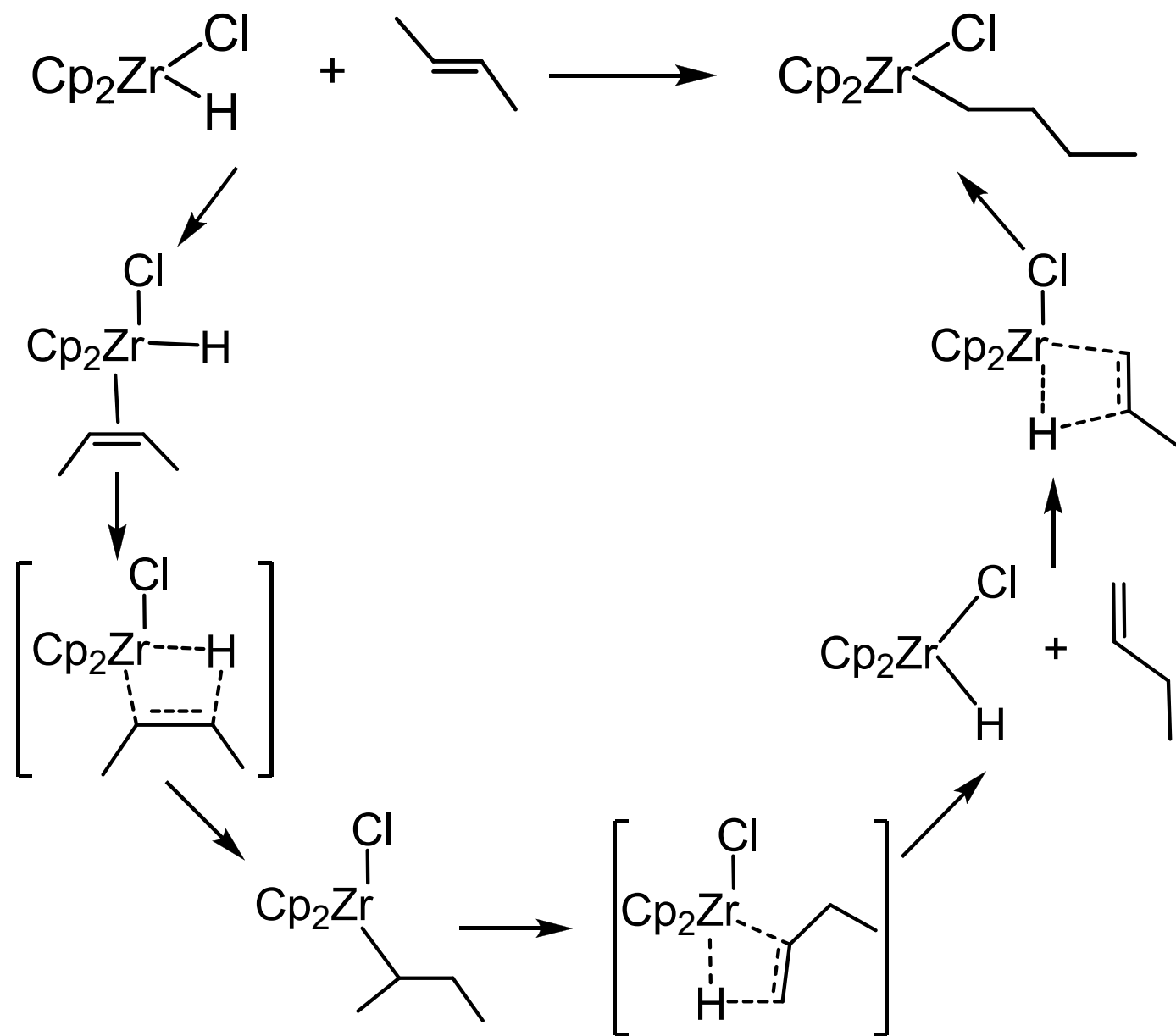
Q1 Propose a mechanism for the following reaction:



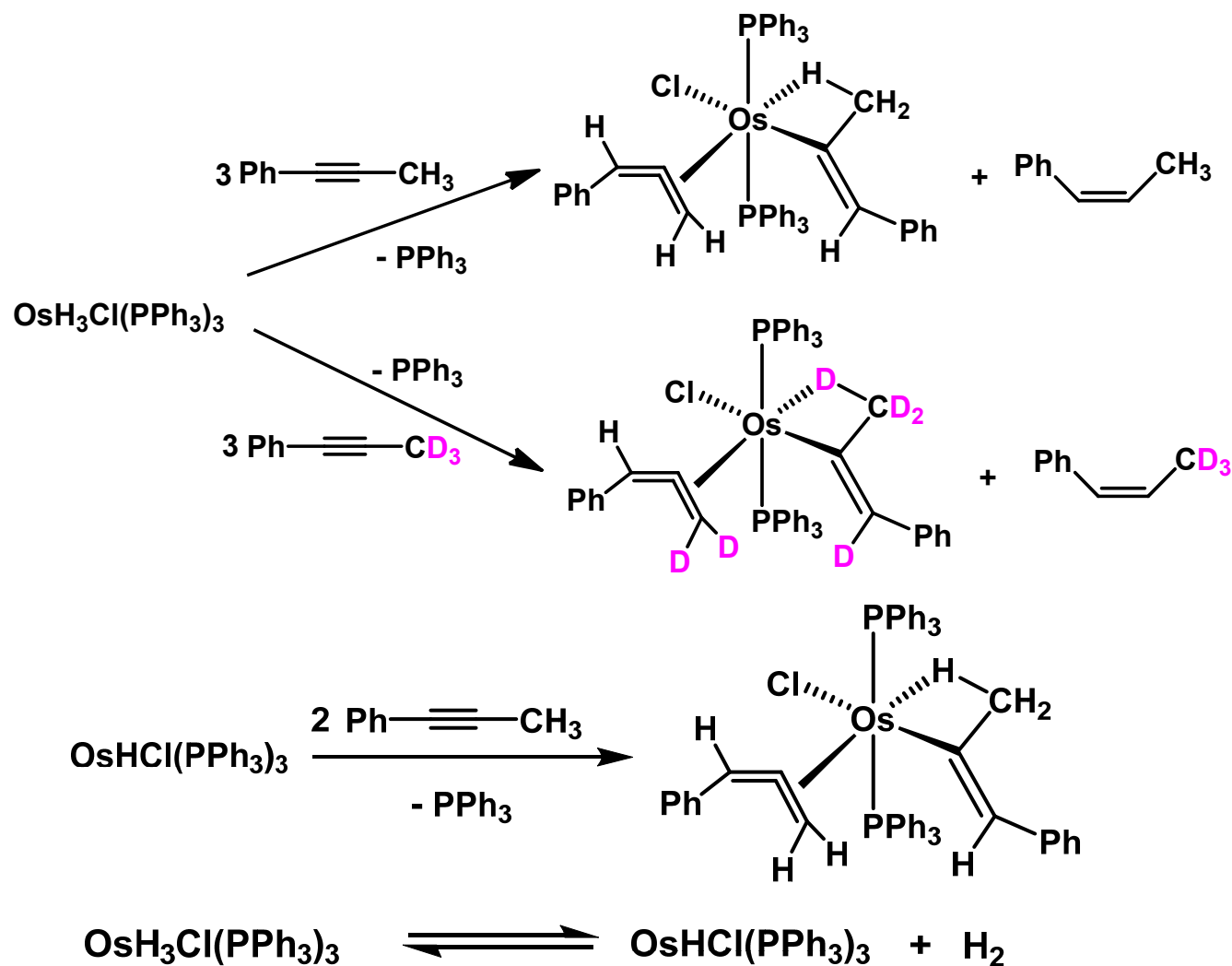
Q1



Q2 Propose a mechanism for the following reaction:



Isomerization of $\text{CH}_3\text{C}\equiv\text{CPh}$ to Phenylallene Promoted by an Osmium Hydride Complex



(Wen, T. B., Jia, G. *et al*, *Organometallics*, **2000**, 19, 3466.)

Proposed mechanism for the formation of allene

